Introduction to Machine Learning



Contents from Machine Learning by Sebastian Raschka

Picked up Notebooks

- EDA: การวิเคราะห์ความเสี่ยงการกู้เงิน
- https://www.kaggle.com/janiobachmann/lending-club-risk-analysis-and-metrics/notebook



Picked up Notebooks

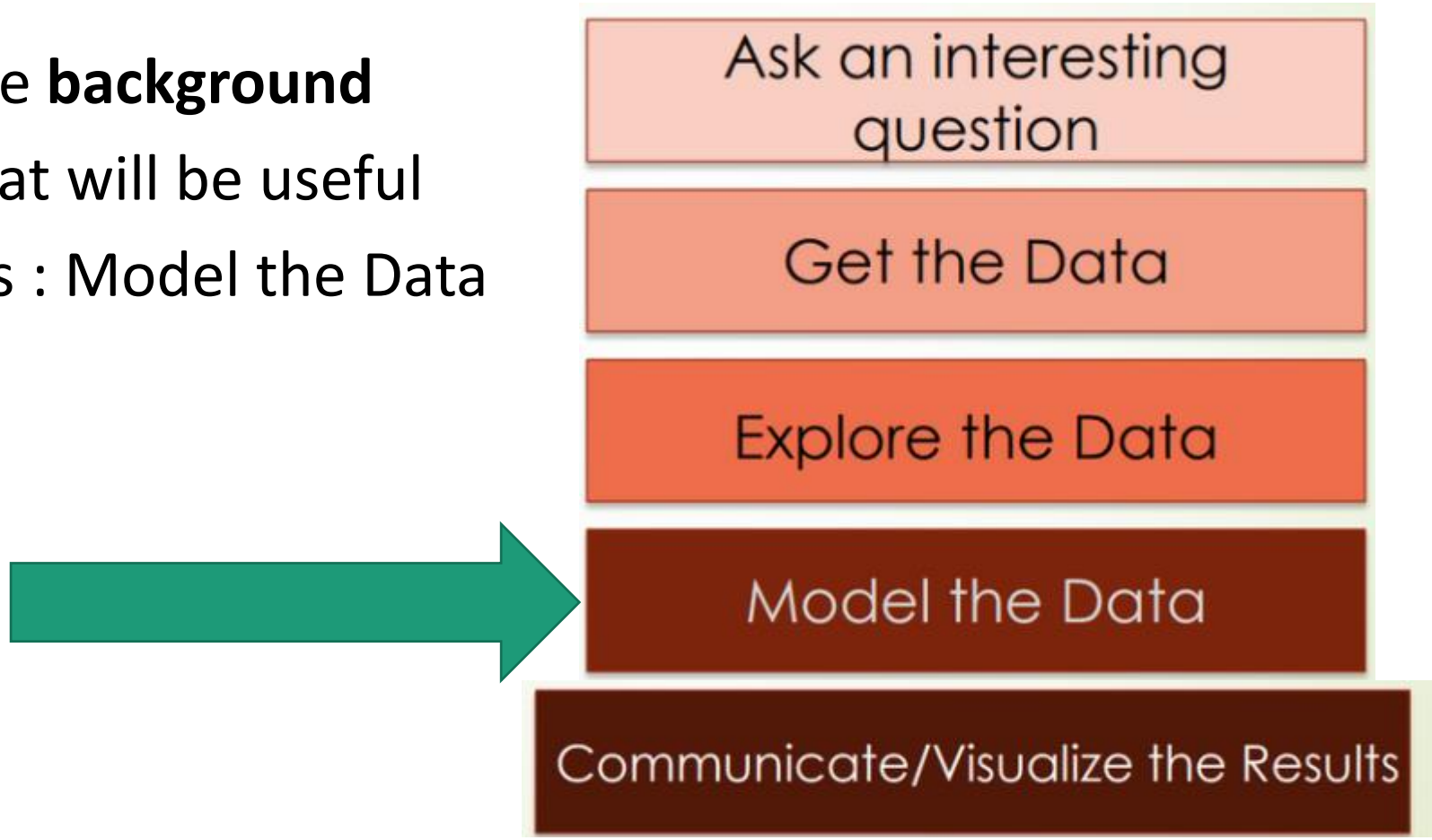
- ML Pipeline framework
- https://www.kaggle.com/ldfreeman3/a-data-science-framework-to-achieve-99-accuracy

Picked up Notebooks

- Clustering Customer Segmentation
- https://www.kaggle.com/yasirhussain1987/customer-segmentation-using-k-means/

Recap Data Science Process

- Today will be about the **background of Machine Learning** that will be useful for Data Science Process : Model the Data



Outline

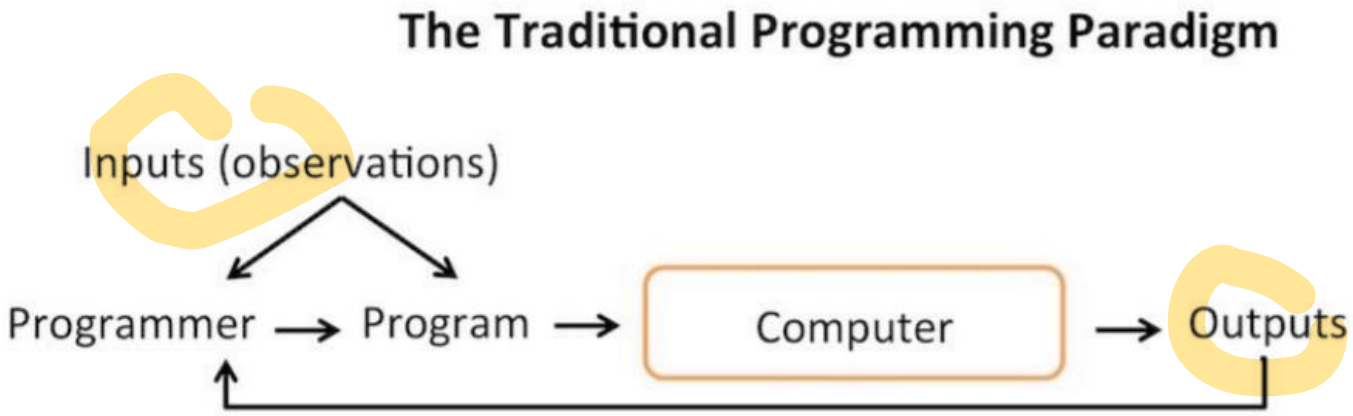
- **What Machine Learning is**
- Categories of Machine Learning
  - Supervised Learning
  - Unsupervised Learning
  - Reinforcement Learning
- **Machine Learning Life Cycle**
  - **Supervised Learning Workflow**
- Basic Steps for Applied Machine Learning (ML)
- Machine Learning Algorithms

What Machine Learning is

"Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed"

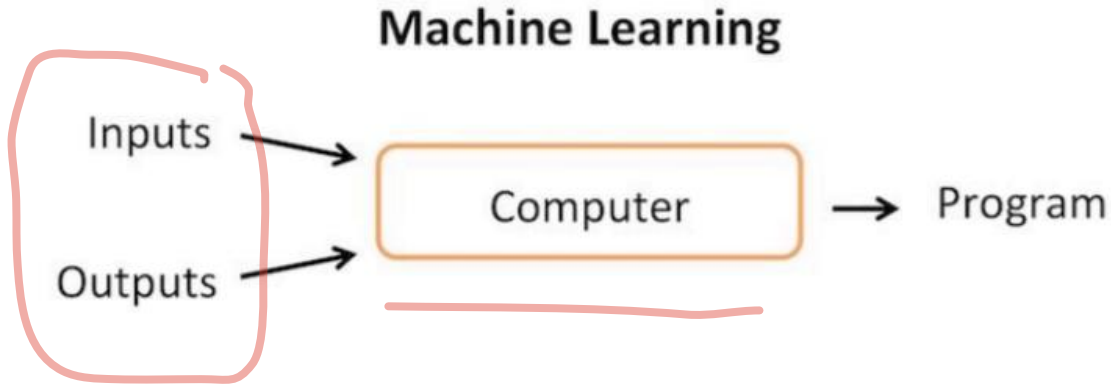
— Arthur L. Samuel, AI pioneer, 1959

Machine Learning: Programming Paradigm





https://www.pinterest.com/pin/766315692821462168/visual-search/?cropSource=6&h=297&w=530&x=17&y=10



Some Applications of Machine Learning in Data Science

- **Finance**
  - **Fraud and Risk Detection**
  - **Stock market prediction**

- **Customer Support Queries (and Chatbots)**
  - **Virtual assistance for patients and customer support**

- **Medical Image Analysis**
  - **Functional network connectivity**

- **Recommendation**
  - **LinkedIn and Facebook recommendations and ads**
  - **App Store and Play Store Recommendations**
  - **Targeted Advertising**

- **Natural Language Processing**
  - **Sentimental Analysis**
  - **Language translation**

Image recognition: Deep Learning

Image recognition for quality assurance

- Image analysis of equipment, inspect characteristics such as e.g. corrosion, cracks, weld quality etc...

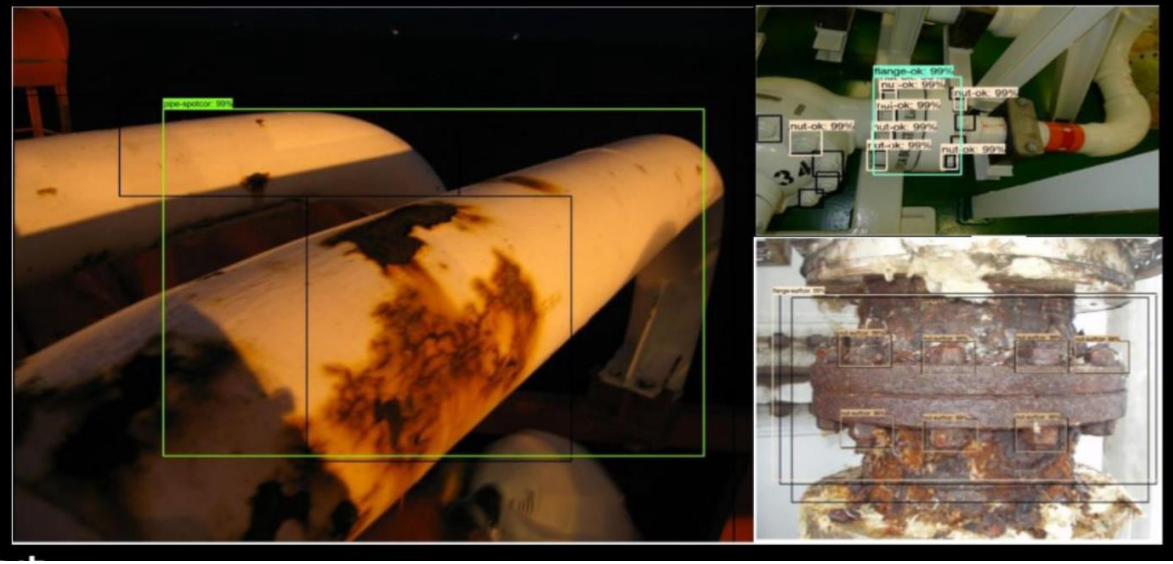


Image recognition in aquaculture, fish health

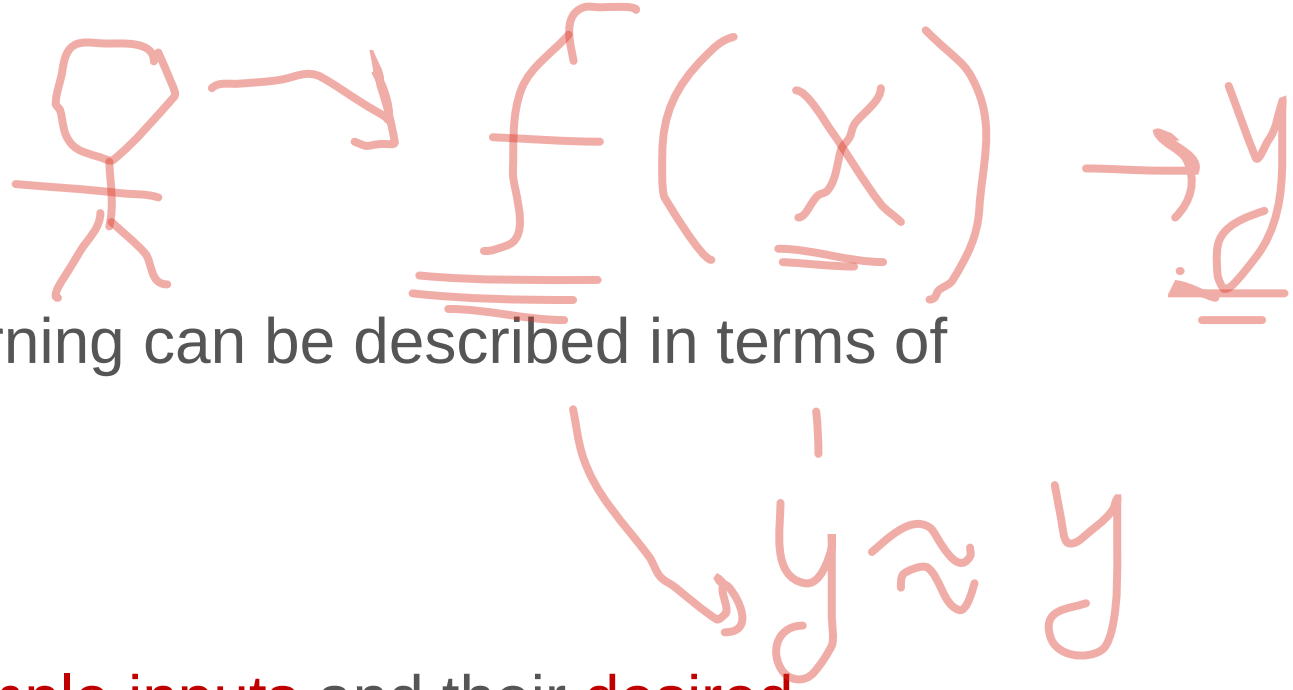
- Automatic image analysis, extracting information on lice, decease, deformities, ++
- Allows for real-time monitoring of fish health on large fish farms



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Supervised Learning



"Supervised learning in machine learning can be described in terms of function approximation."

In other words,

The computer is presented with example inputs and their desired outputs, given by a "teacher", and the goal is to learn a general rule that maps inputs to outputs.



What would be an approximation for $f(x)$?

X = ขนาดพื้นที่ (ตารางเมตร)	10	20	35	45	60	80
Y = ราคา	10000	20000	35000	45000	60000	?

$f(x) \approx 1000x$ (Parametric)

TASK: Use the pattern in my data to make predictions - ***predictive***

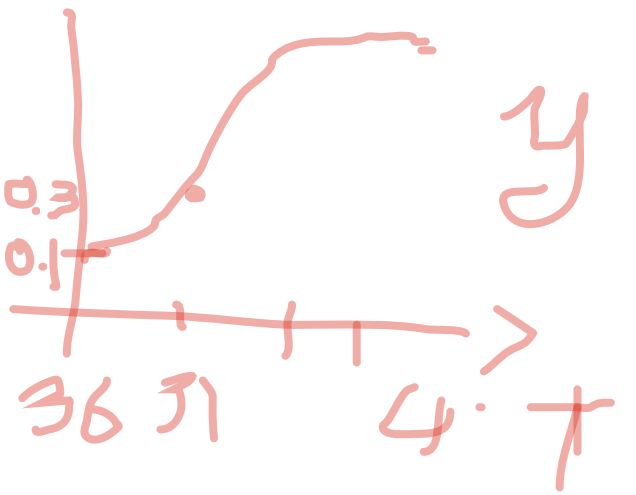
What would be an approximation for $f(x)$?

X = ขนาดพื้นที่ (ตารางเมตร)	**10**	**30**	**35**	**40**	**60**	**80**
Y = ราคา	10000	20000	?	45000	60000	70000

$f(x) \approx$ ค่าเฉลี่ยราคาของขนาดพื้นที่ห้องที่ใกล้เคียง x (Lazy Learner, Non-Parametric)

What would be an approximation for $f(x)$?

Name	อุณหภูมิร่างกาย	เพิ่งเดินทางกลับจากต่างประเทศ	สัมผัสผู้มีประวัติติดเชื้อ	กลุ่มเสี่ยง
001	36	Yes	Yes	มาก
002	39	No	No	น้อย
003	37	No	Yes	มาก
004	37	Yes	No	ปานกลาง
005	39	Yes	No	?

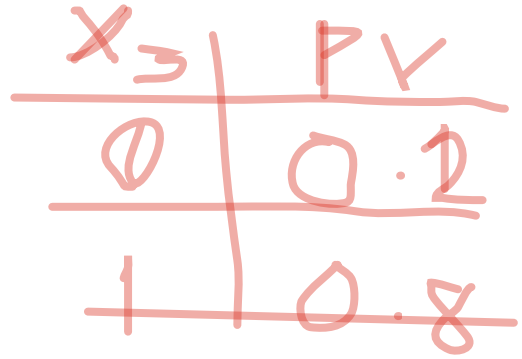


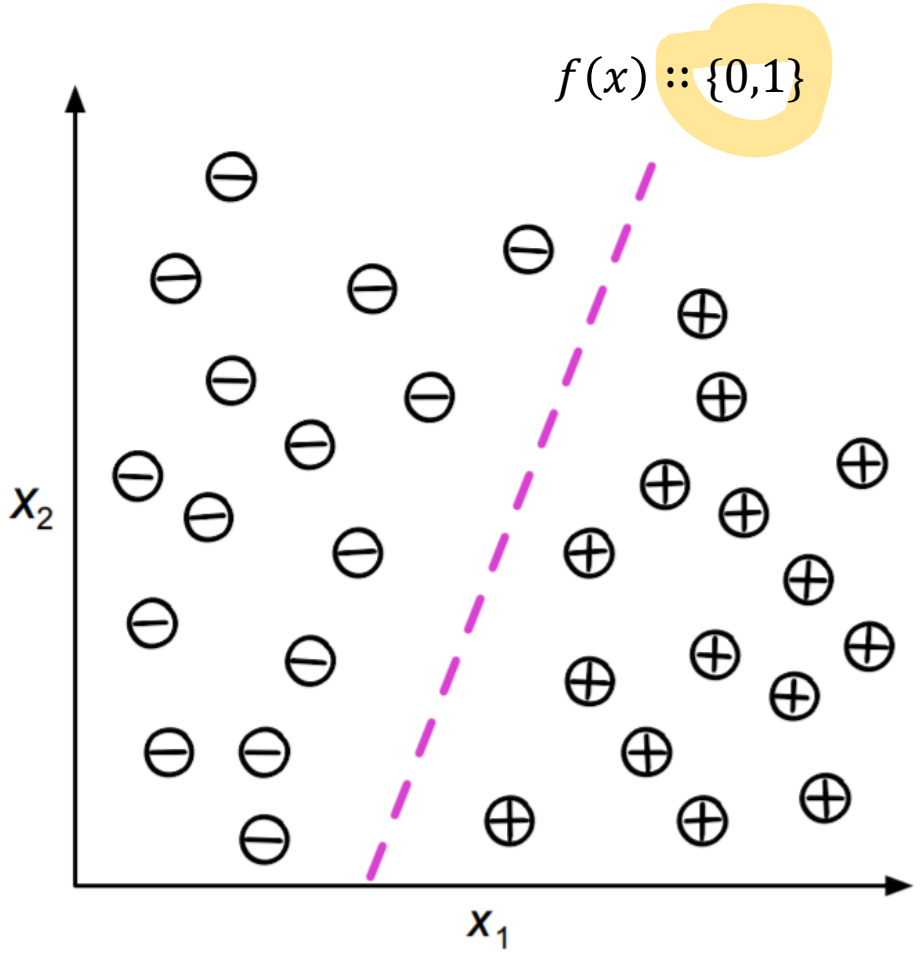
$y = f(39, Yes, No)$

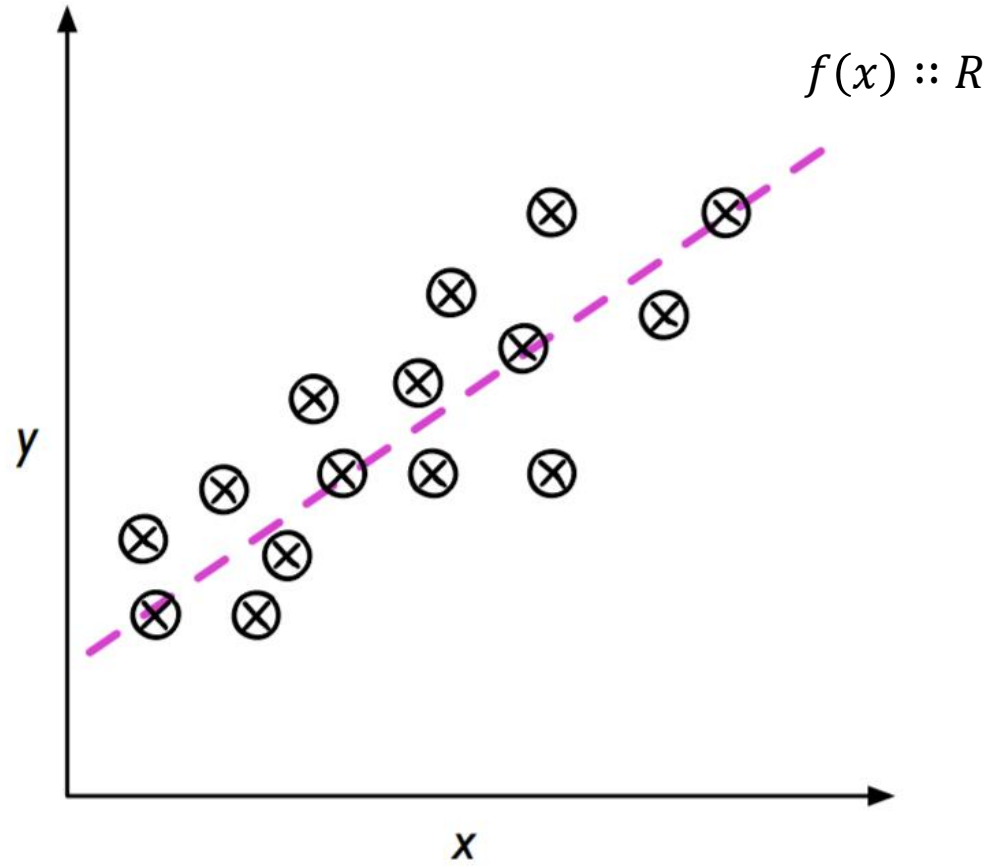
$= \ldots 0.1 + 0.9 + 0.2$

X_2	Pr
0	0.1
1	0.9

Type of Supervised Learning: Classification vs Regression



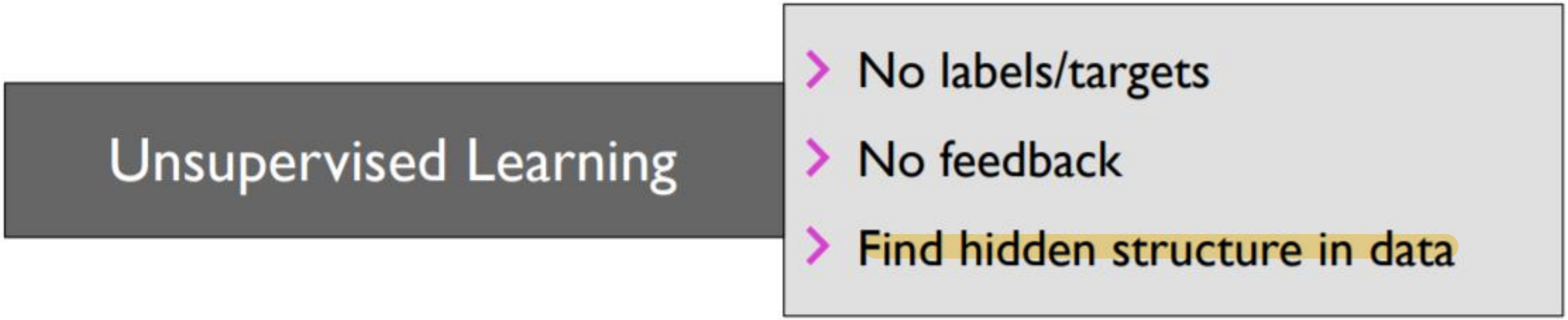




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  - Supervised Learning
  - **Unsupervised Learning**
  - Reinforcement Learning
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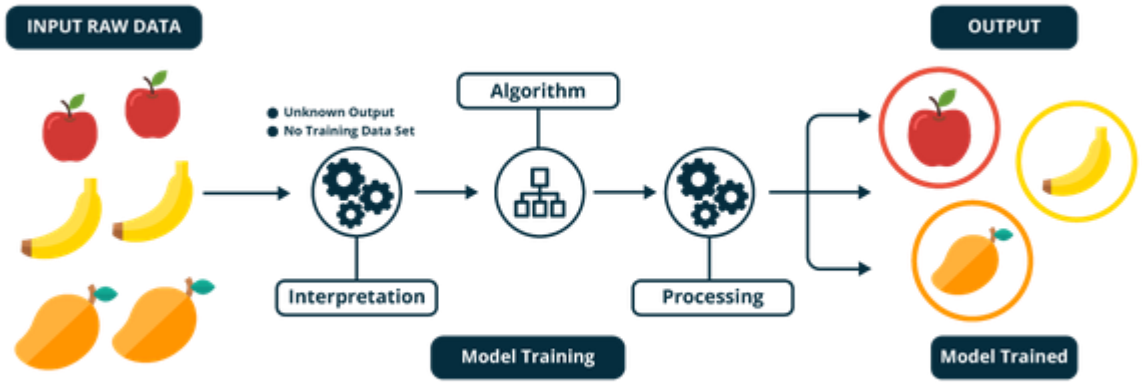
Unsupervised Learning:



TASK: Tell me what pattern exist in my DATA - descriptive

Fig: https://www.onemodel.co/blog/ai-academy-what-is-machine-learning

Unsupervised Learning: Clustering



WHAT do we have to tell the clustering algorithm, so it can group similar objects together?

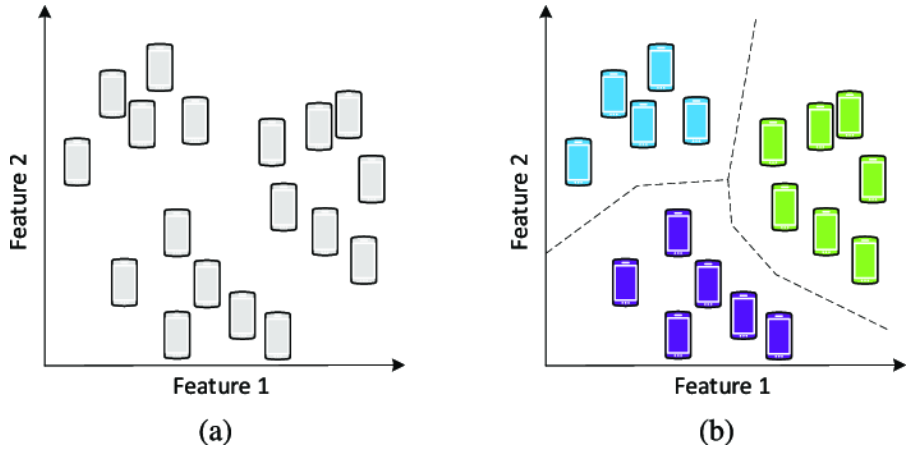
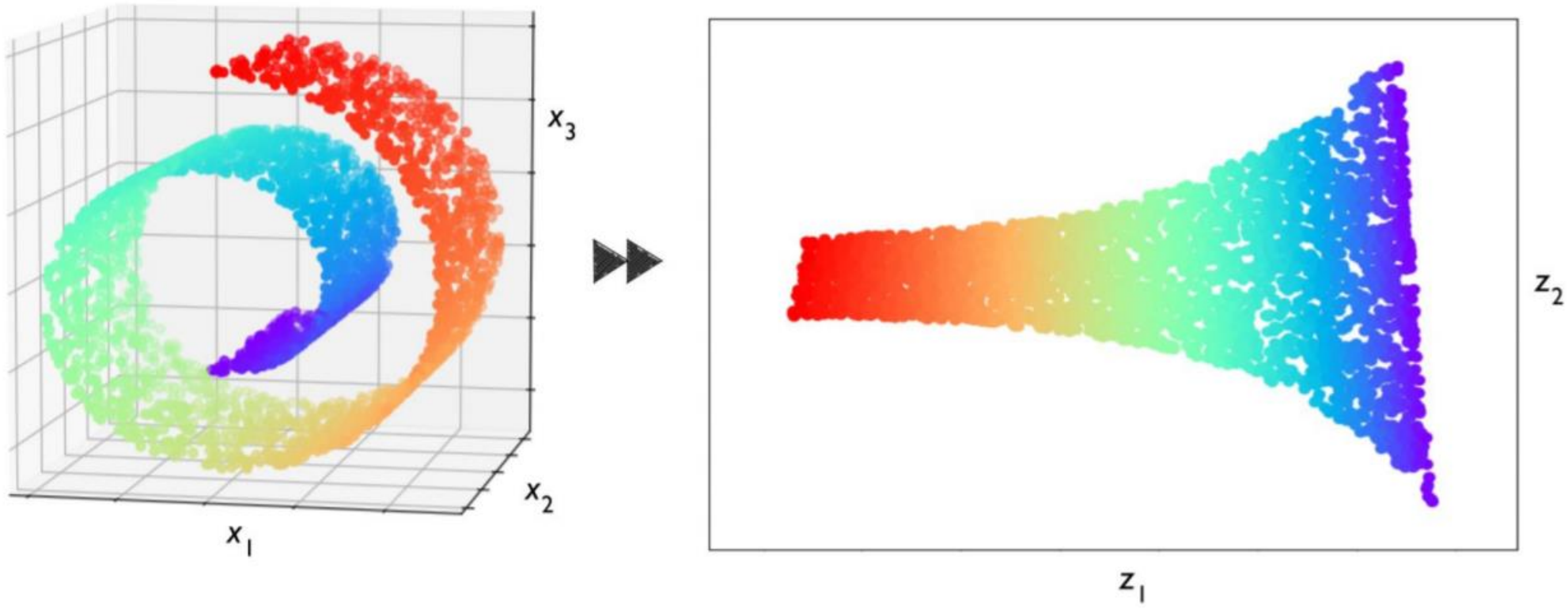


Fig: https://www.researchgate.net/figure/Unsupervised-learning-example-a-Initial-set-of-unlabeled-data-with-different-user_fig6_335937022

Unsupervised Learning: Dimensional reduction



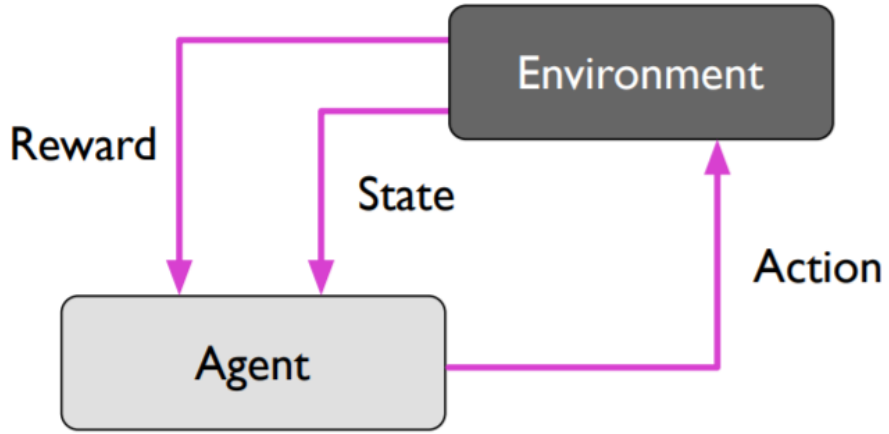
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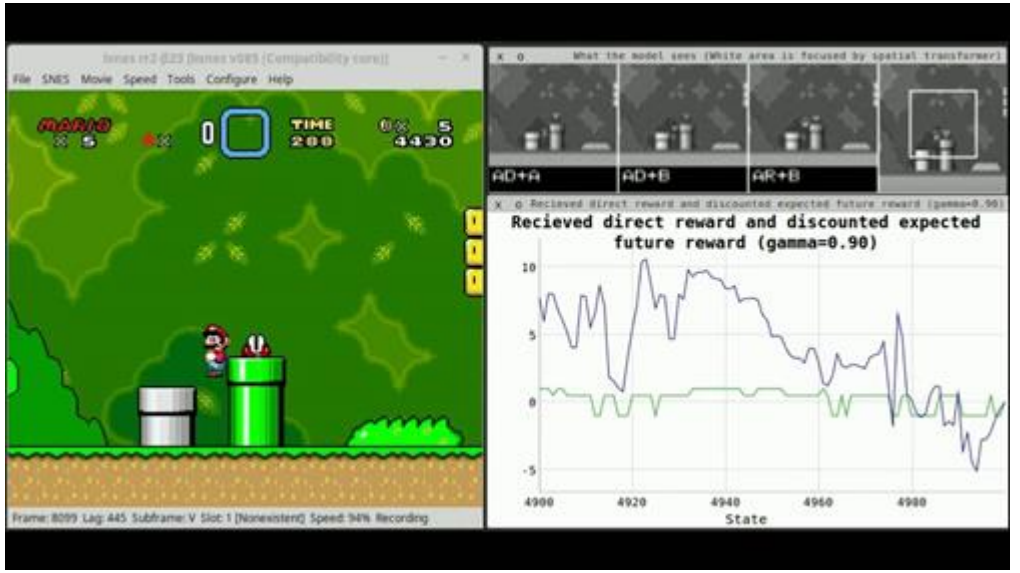
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Reinforcement Learning

Reinforcement Learning

- Decision process
- Reward system
- Learn series of actions



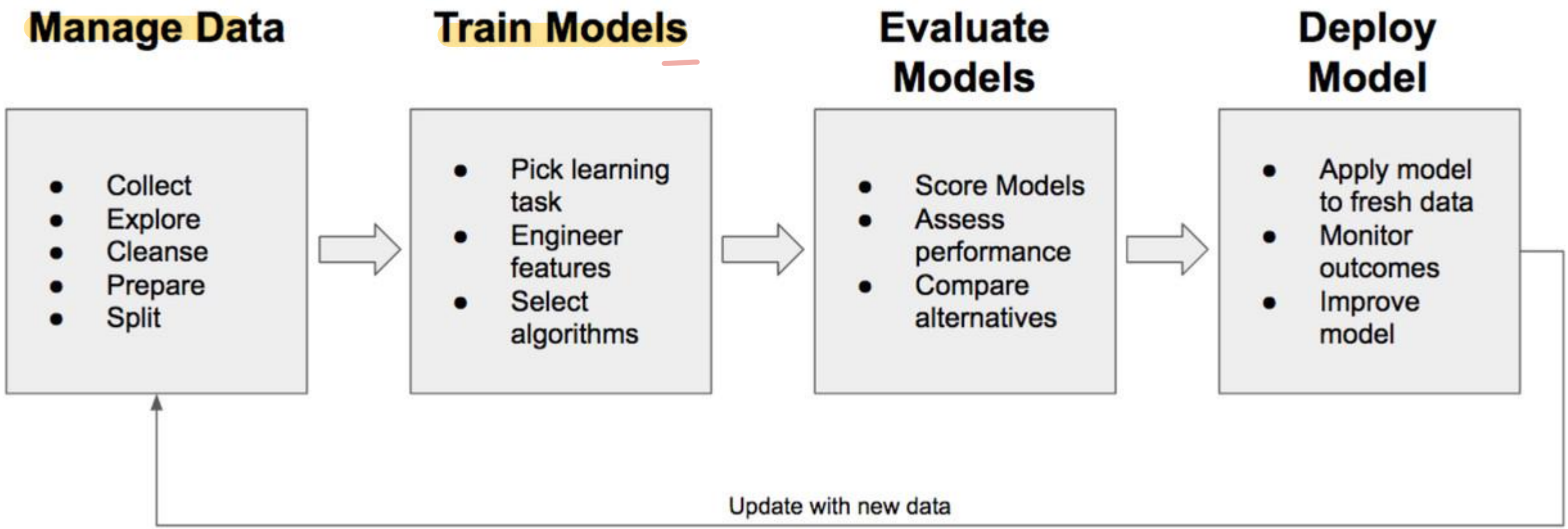


https://medium.com/@datasciencemilan/reinforcement-learning-overview-7f3ceba0bb60

Outline

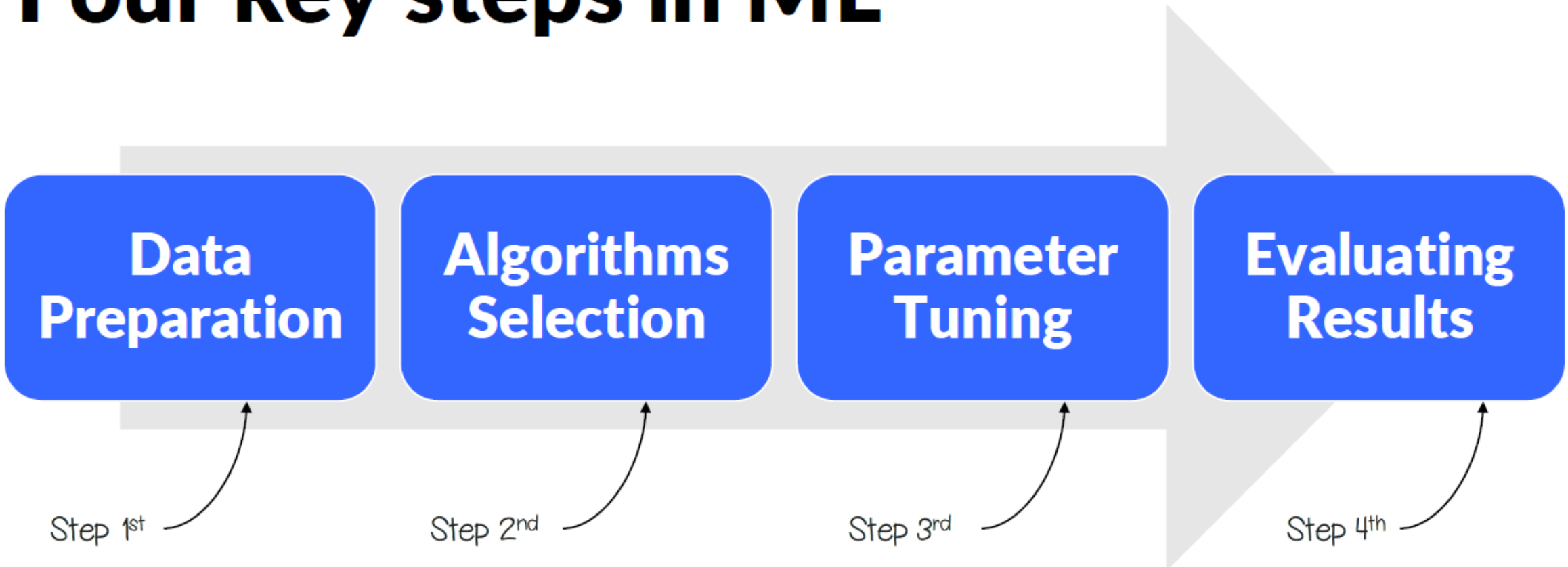
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Machine Learning Modeling Cycle



https://www.linkedin.com/pulse/4-stages-machine-learning-ml-modeling-cycle-maurice-chang/

Four key steps in ML



Algorithms Selection

- The model error can be decomposed into three sources of error: the **variance** of the model, the **bias** of the model, and the variance of the **irreducible error** in the data.

- Error(Model) = Variance(Model) +
  Bias(Model) +
  Variance(Irreducible Error)

Bias and Variance

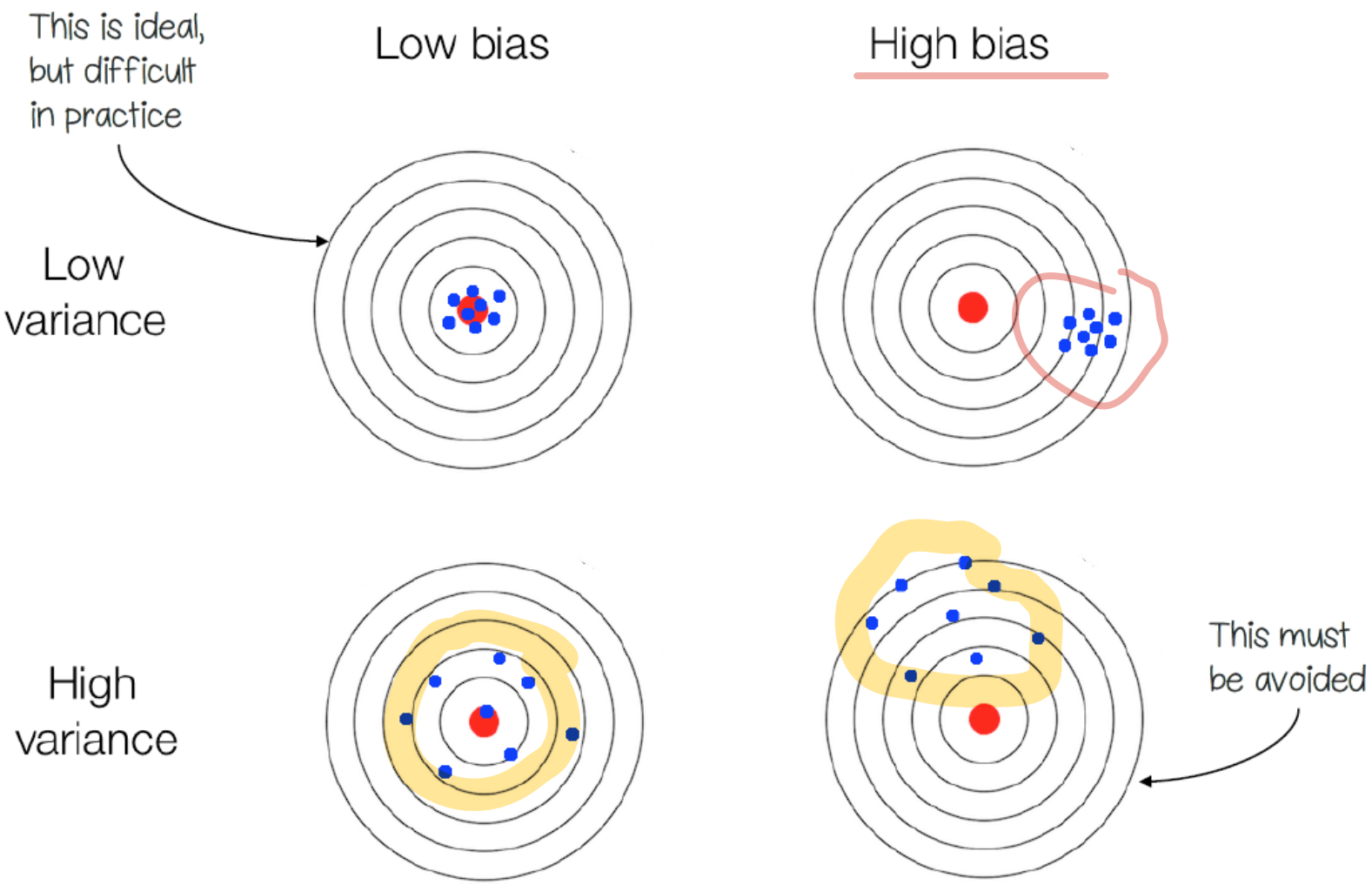
- **Model Bias**

The bias is a measure of how close the model can capture the mapping function between inputs and outputs.

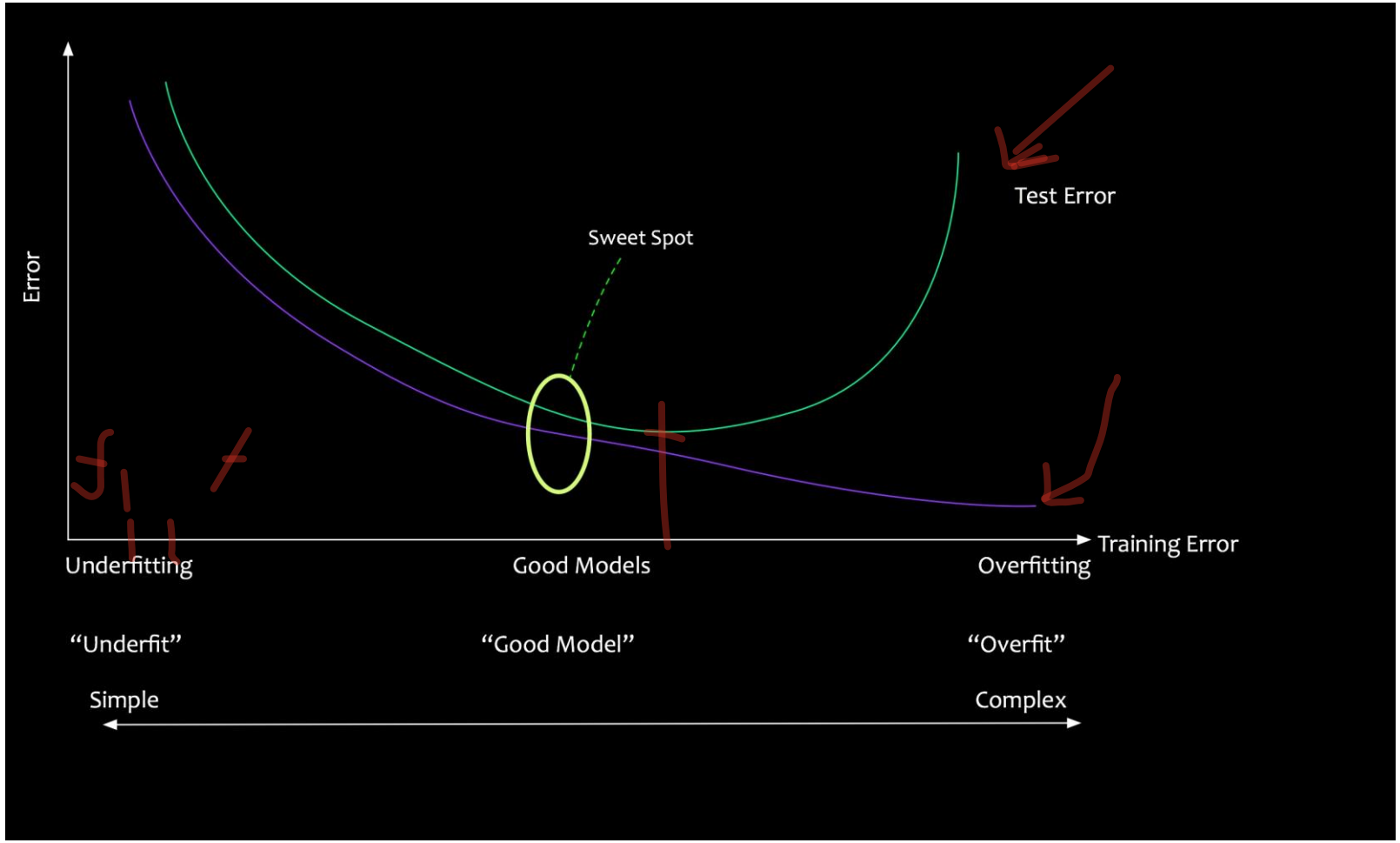
- **Model Variance**

The variance of the model is the amount the performance of the model changes when it is fit on different training data.

Bias vs Variance



Bias / Variance a way to understand overfitting and underfitting

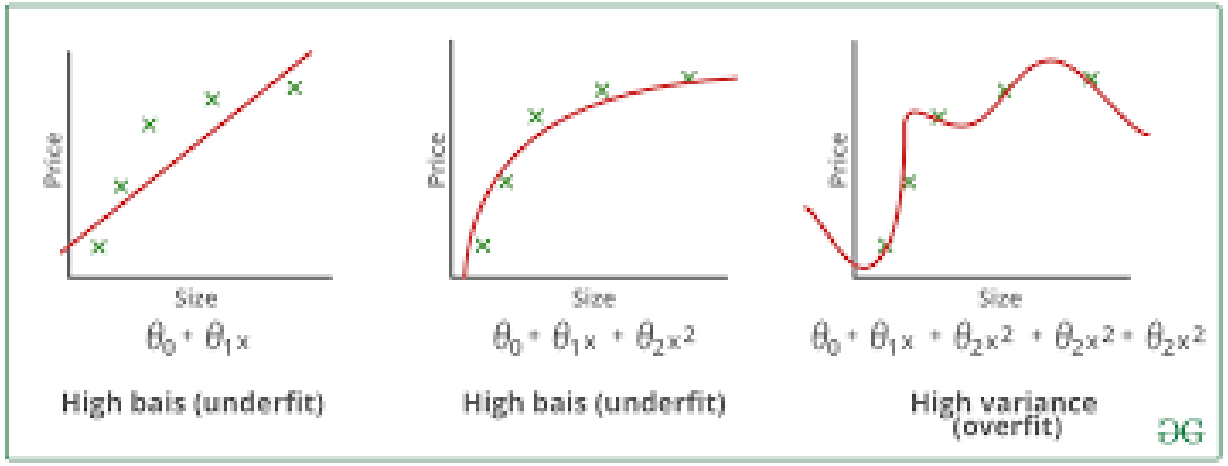


Example Models

High Bias *Having lots of assumptions to do function approximation*	*Linear Regression* *Logistic Regression* *Linear Discriminant Analysis*	***Low Variance*** *New data does not cause model to change much*
Low Bias *Having little or few assumptions about the function approximation*	*Decision Tree* *KNN* *SVM*	***High Variance*** *New data causes model to change much i.e. prediction results vary significantly compared to low variance algorithms*

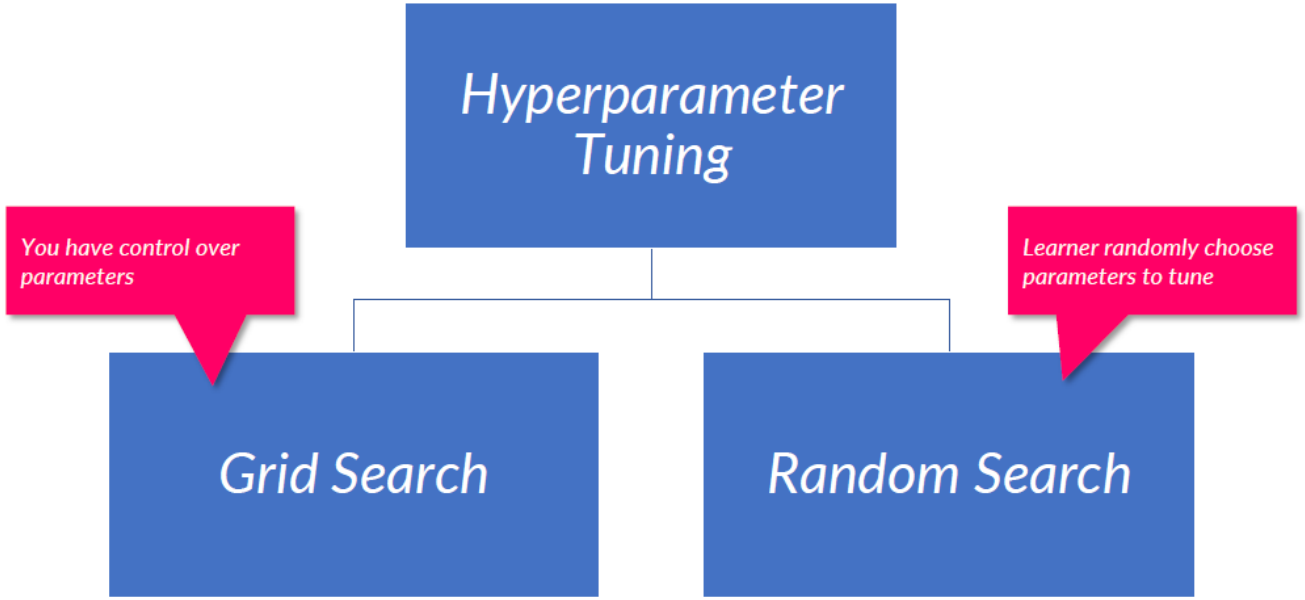
Image Taken from Introduction to Machine Learning by Datarocky

Parameter Tuning



SEARCH for the best tune

ML is optimization problem

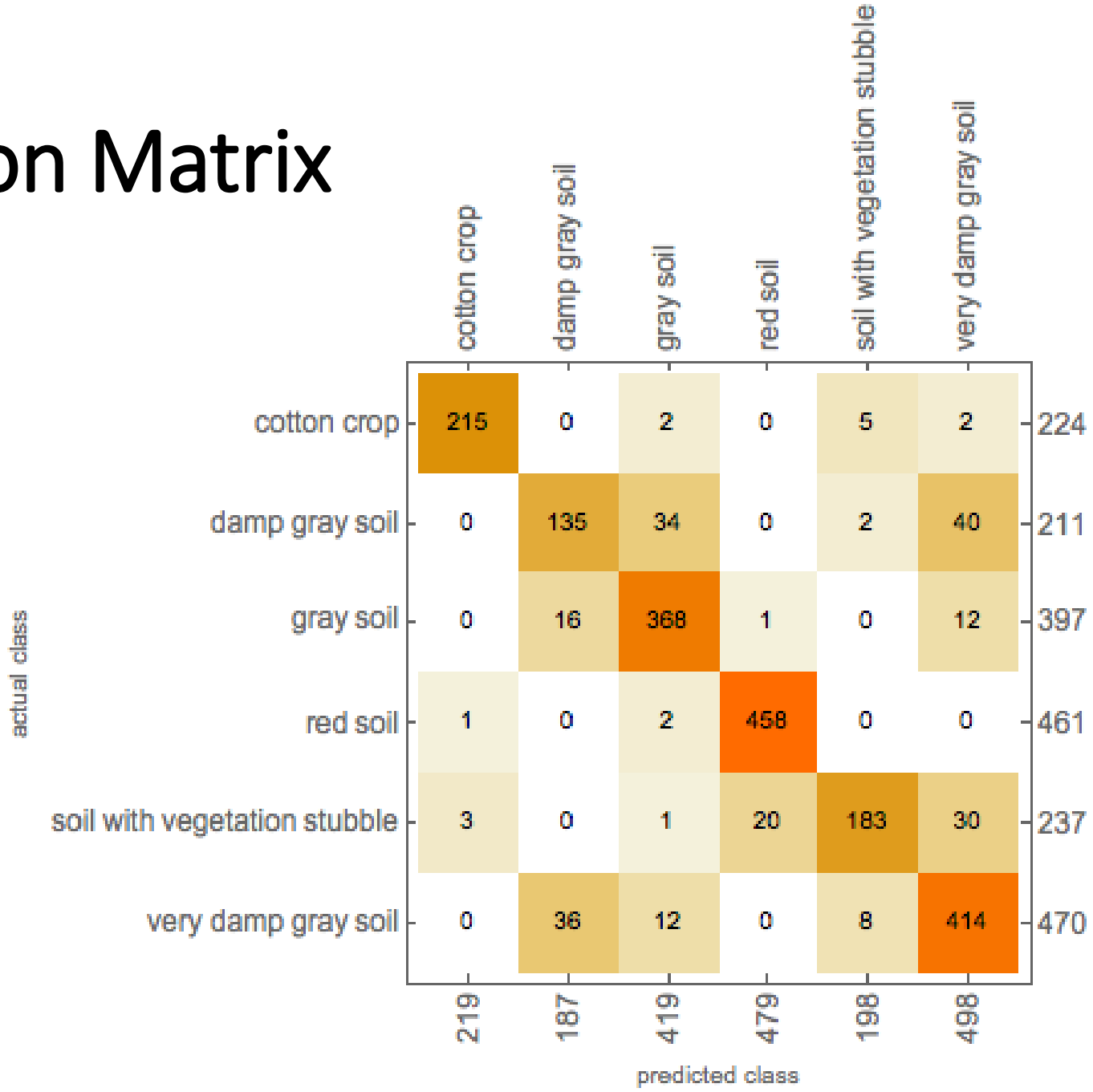


Performance Metrics

- Classification problems
  - Accuracy percentage
  - Confusion Matrix
  - Area Under the Curve (AUC)

- Regression problems
  - Root Mean Square Error (RMSE)
  - Mean Absolute Error (MAE)

Confusion Matrix

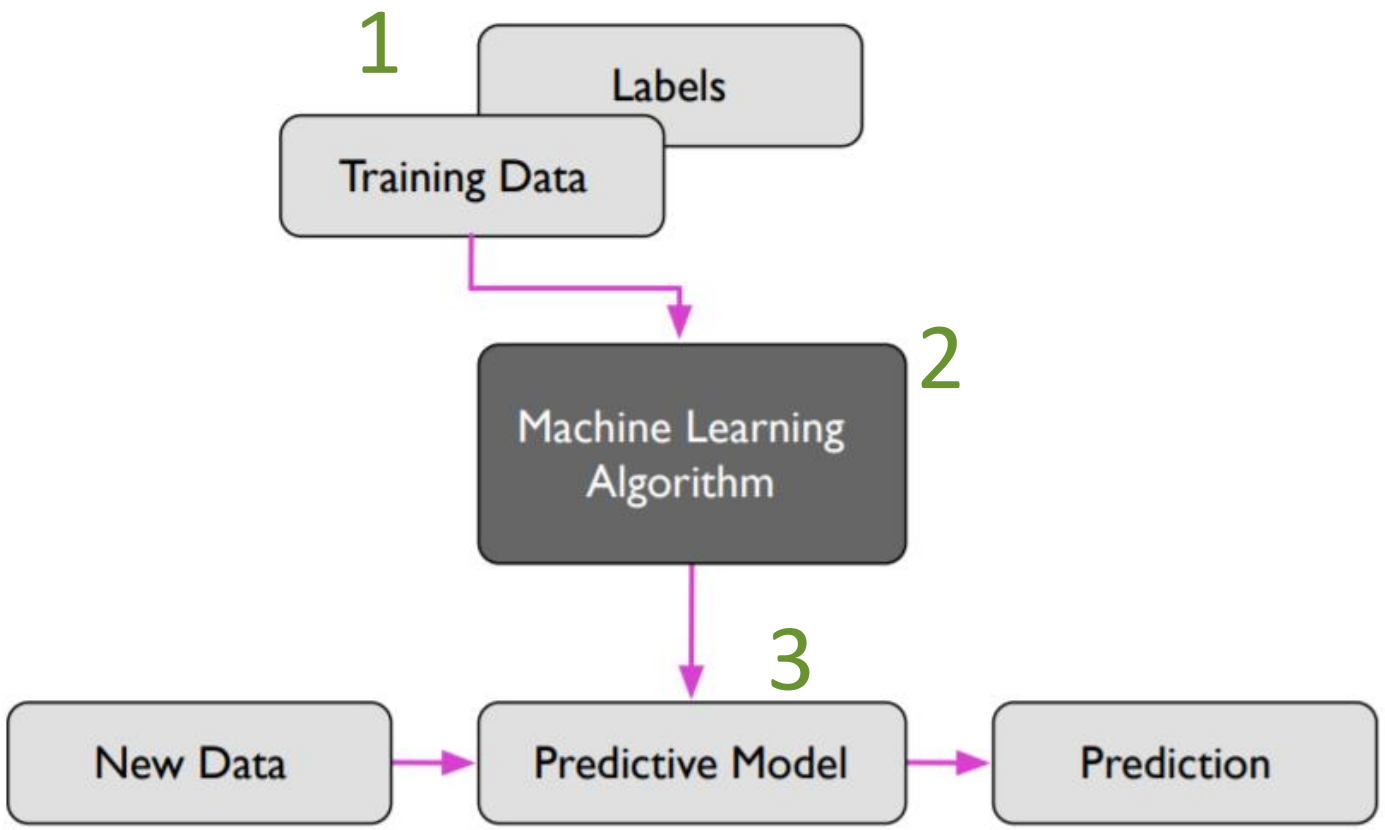


https://www.wolfram.com/mathematica/new-in-10/highly-automated-machine-learning/visualize-the-confusion-matrix-of-a-classifier.html

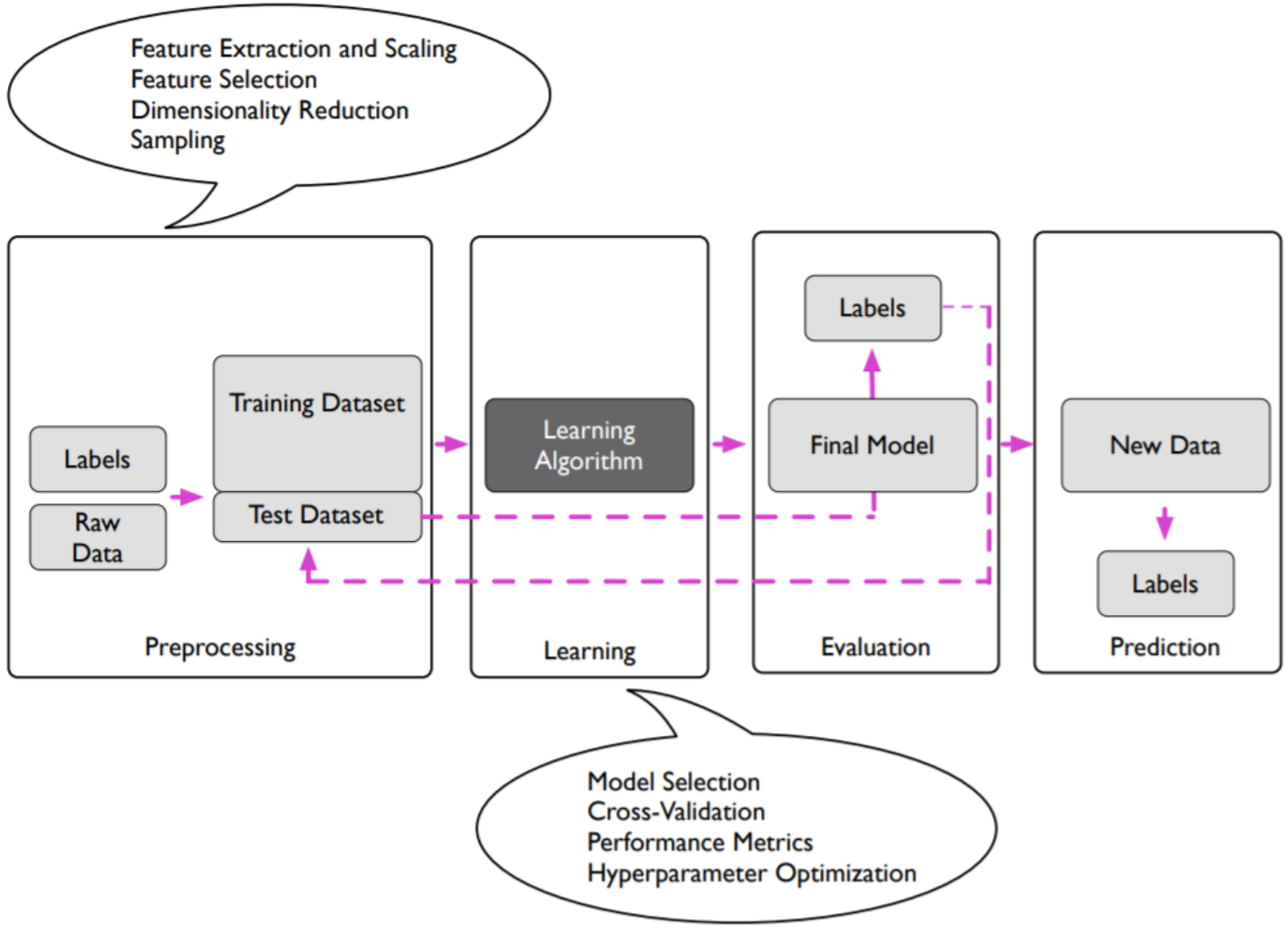
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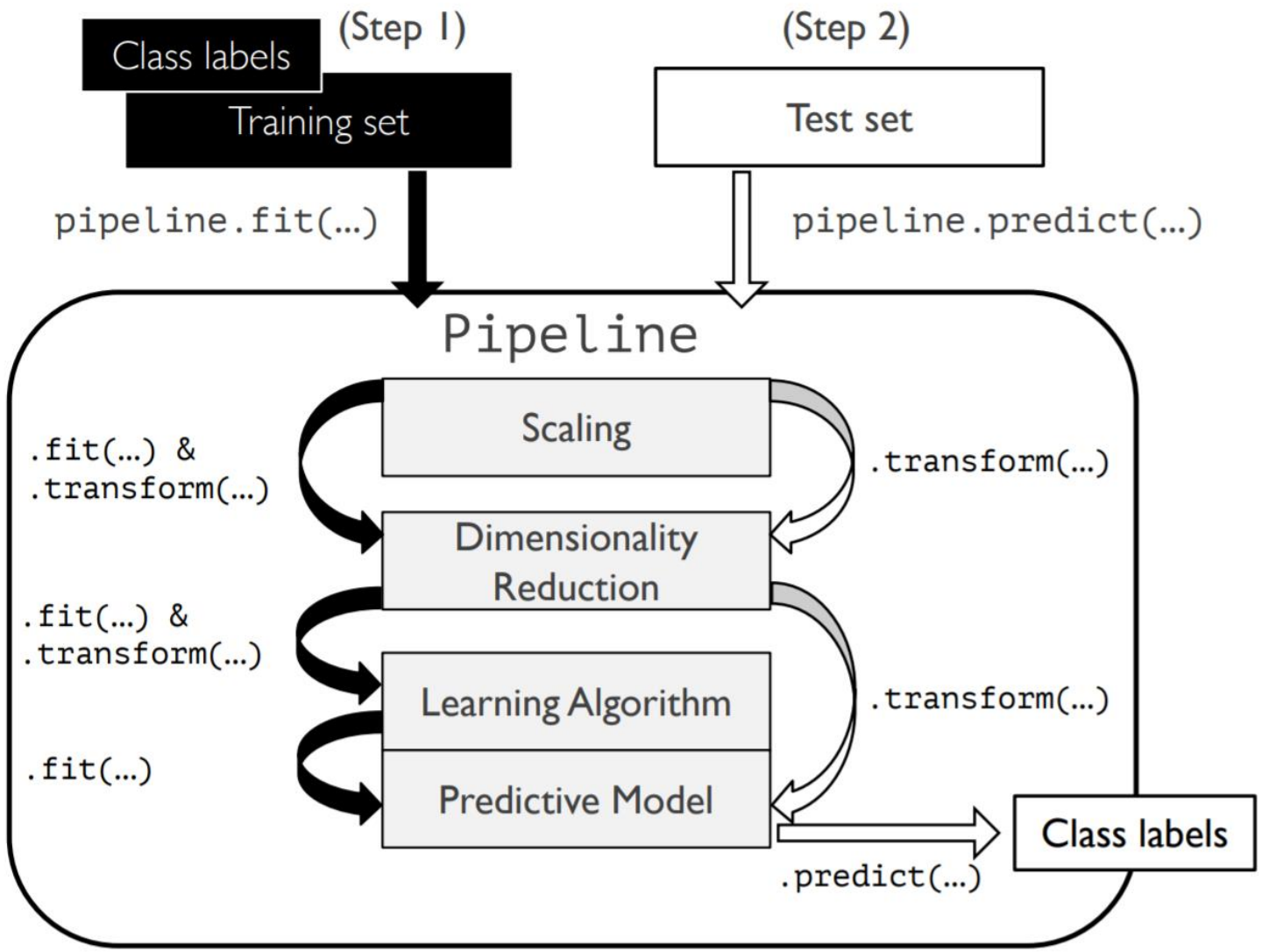
Supervised Learning Workflow



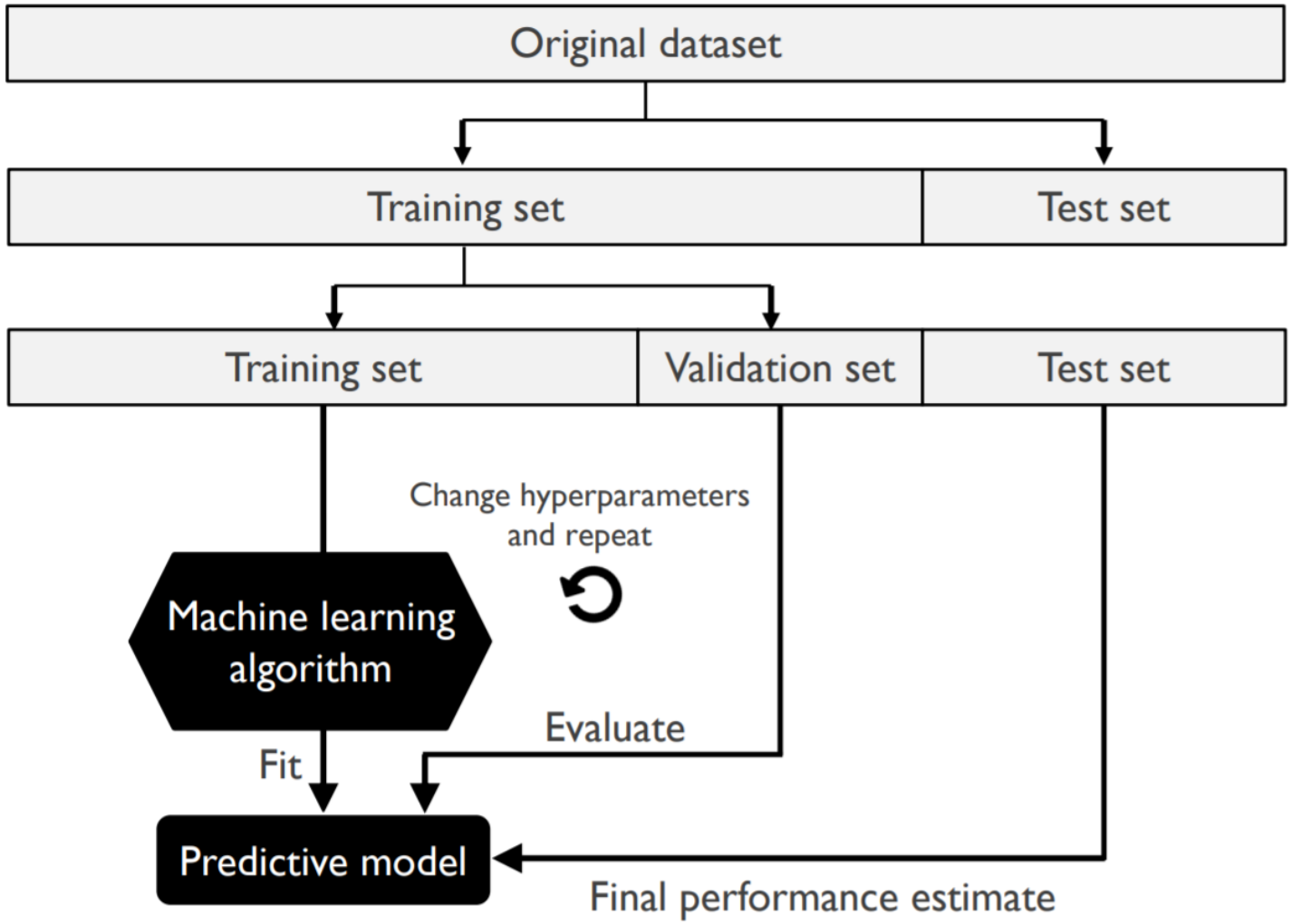
Supervised Learning Workflow



Scikit-Learn Pipelines



Performance Evaluation



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Basic Steps for Applied Machine Learning

- Understand the Problem Statement & Import Packages and Datasets.
- Perform EDA (Exploratory Data Analysis) - Understanding the Datasets:
  - Explore Train and Test Data and get to know what each Column / Feature denotes.
  - Check for Imbalance of Target Column in Datasets.
  - Visualize Count Plots & Unique Values to infer from Datasets.
- Remove Duplicate Rows from Train Data if present.
- Fill/Impute Missing Values Continuous - Mean/Median/Any Specific Value & Categorical - Others/ForwardFill/BackFill.

Basic Steps for Applied Machine Learning

- Feature Engineering
  - Feature Selection - Selection of Most Important Existing Features.
  - Feature Creation - Creation of New Feature from the Existing Features.

- Split Train Data into Train and Validation Data with Predictors(Independent) & Target(Dependent).

- Data Encoding - Label Encoding, OneHot Encoding and Data Scaling - MinMaxScaler, StandardScaler, RobustScaler

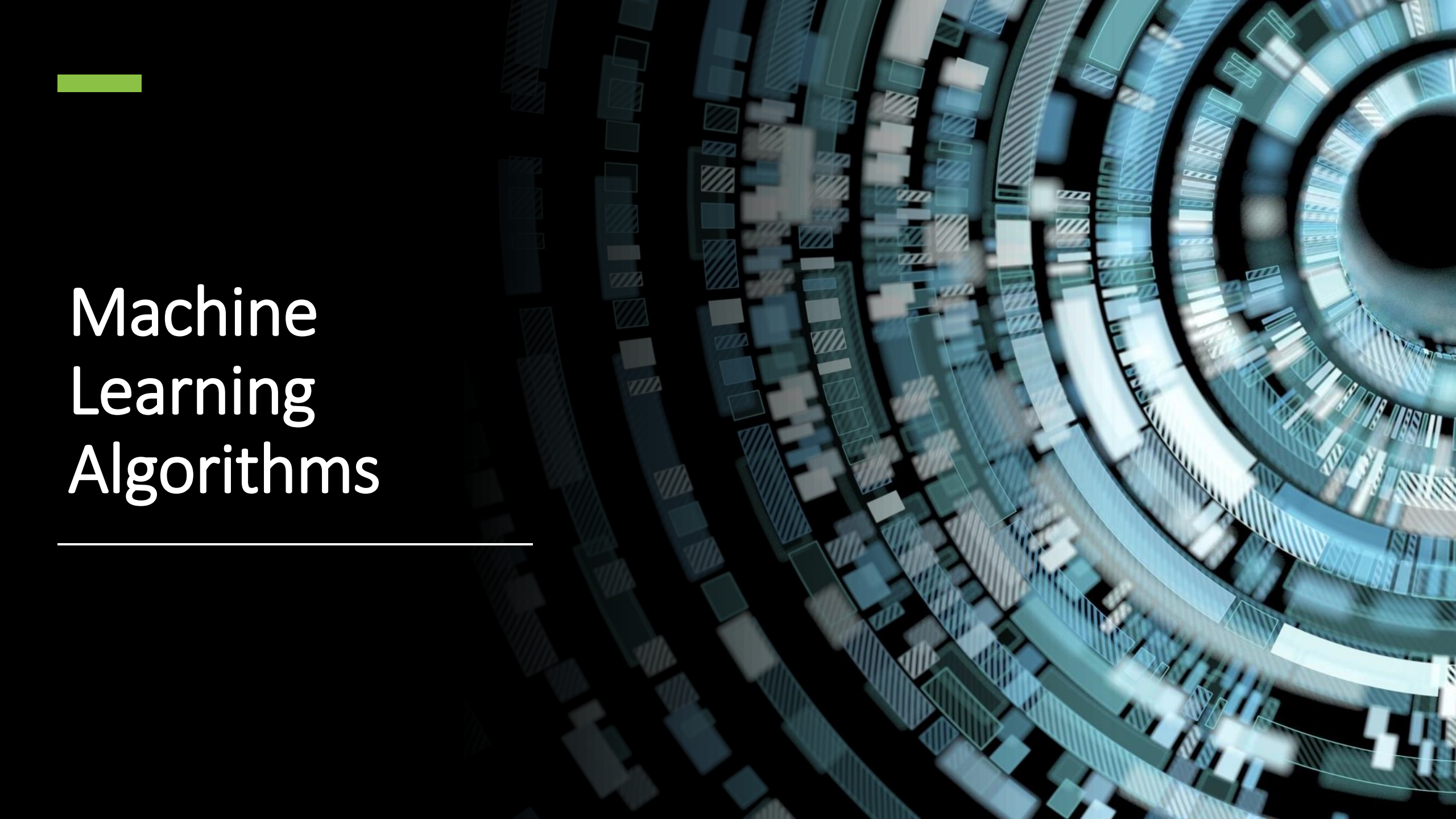
Basic Steps for Applied Machine Learning

- Create Baseline ML Model
- Improve ML Model, Fine Tune with MODEL Evaluation METRIC - "Accuracy" and Predict Target "Outcome“
- Story telling

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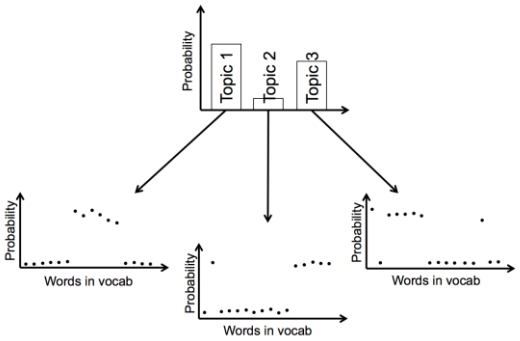
Machine Learning Algorithms



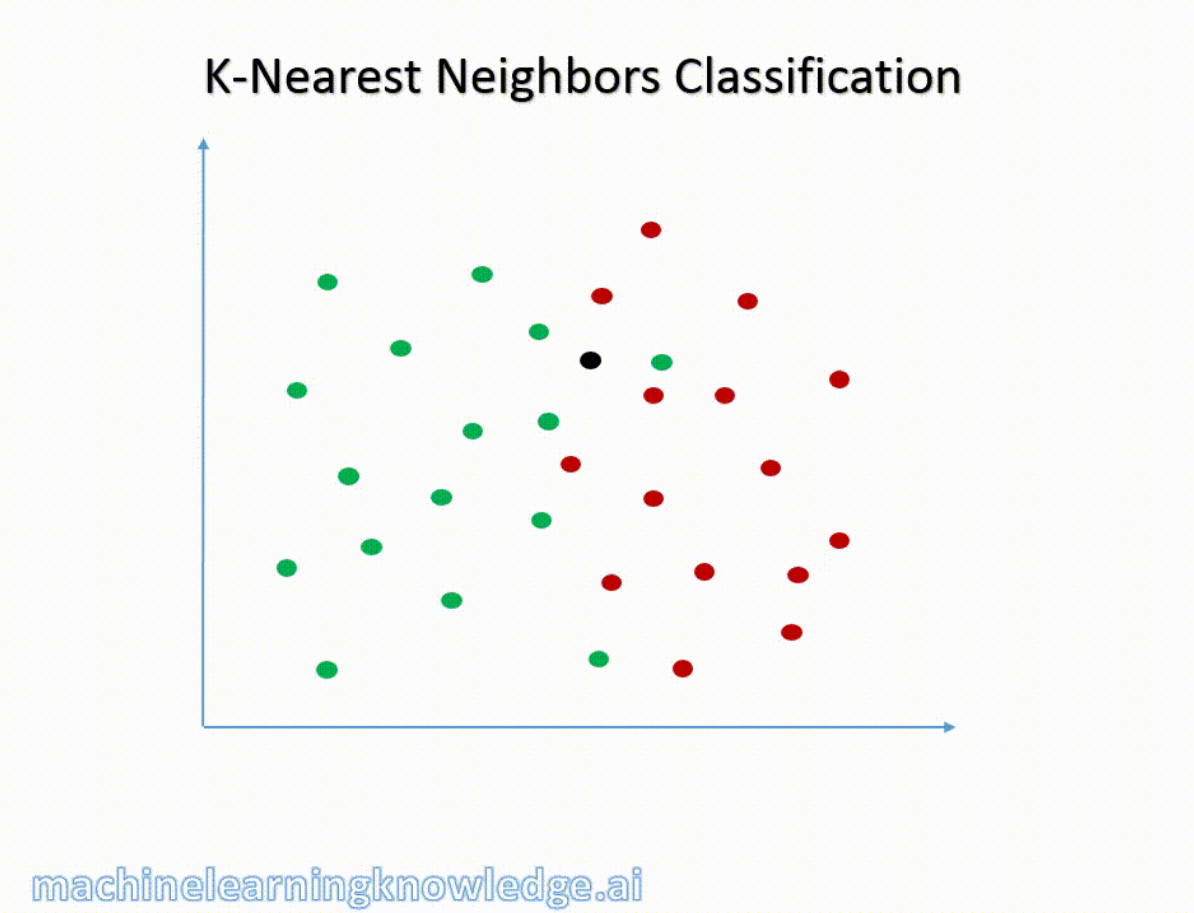
Important Machine Learning Algorithms

- Supervised Learning
  - Classification: K-nn, decision tree, Random Forests
  - Regression: linear regression, logistic regression, Random Forests
  - Dimensional reduction: Linear Discriminant Analysis (LDA)

- Unsupervised Learning
  - Clustering : K-means, Network clustering,
    Latent Dirichlet Allocation (LDA) (document topics)
  - Dimensional reduction : PCA

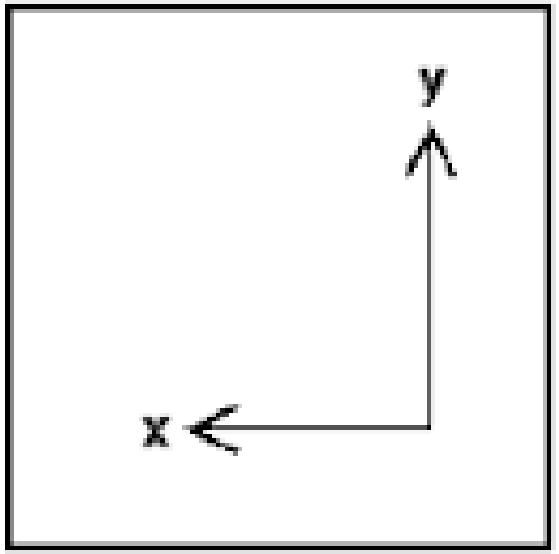


Classification: K-Nearest Neighbors (K-NN)

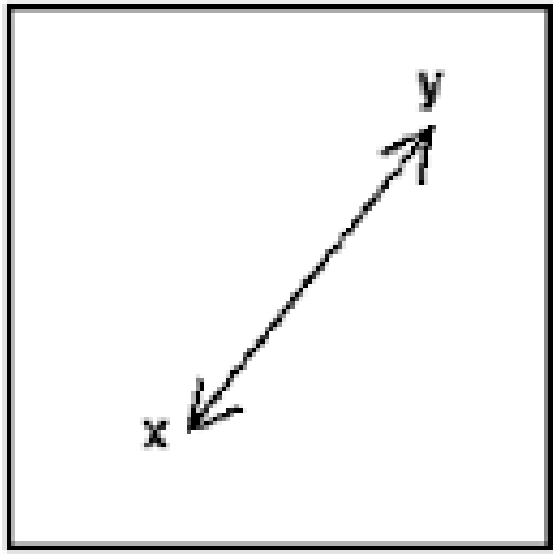


https://machinelearningknowledge.ai/k-nearest-neighbor-classification-simple-explanation-beginners/

Distance function



Manhattan



Euclidean

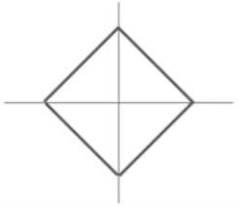
K-Nearest Neighbors: Distance Metric

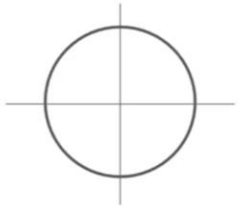
L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

L2 (Euclidean) distance

$$d_2(I_1, I_2) = \sqrt{\sum_p (I_1^p - I_2^p)^2}$$





KNN: Pros and Cons

- Advantage
  - Simple Algorithm
  - No training required (Lazy learner)
  - Ad hock (adding new data not effect the algorithm)

- Disadvantage
  - Does not work well with large dataset
  - Does not work well with high dimensional data
  - Should do feature scaling
  - Sensitive to noisy data, missing value and outliners

K-Nearest Neighbors Classification

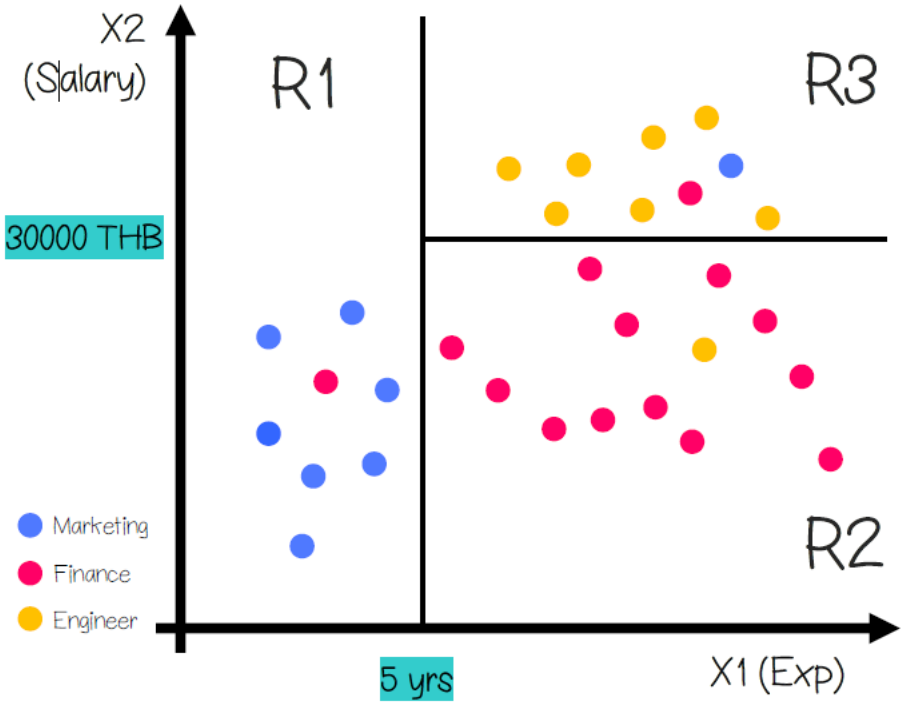
- https://chrisalbon.com/machine_learning/nearest_neighbors/k-nearest_neighbors_classifer/

- COLAB
- https://colab.research.google.com/drive/1W9WKK8MkTho9IUGjTHP8OaCr6QUERLkY?usp=sharing

- https://colab.research.google.com/drive/1TFTftpp1VpqhGfVfwkCCGqkwP_6acCIS?usp=sharing

Decision Tree

- Divide space into regions that best predict data points



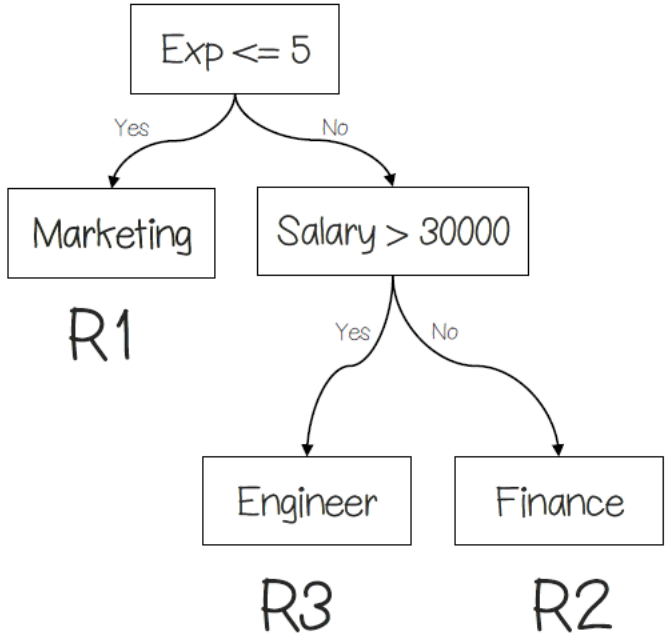
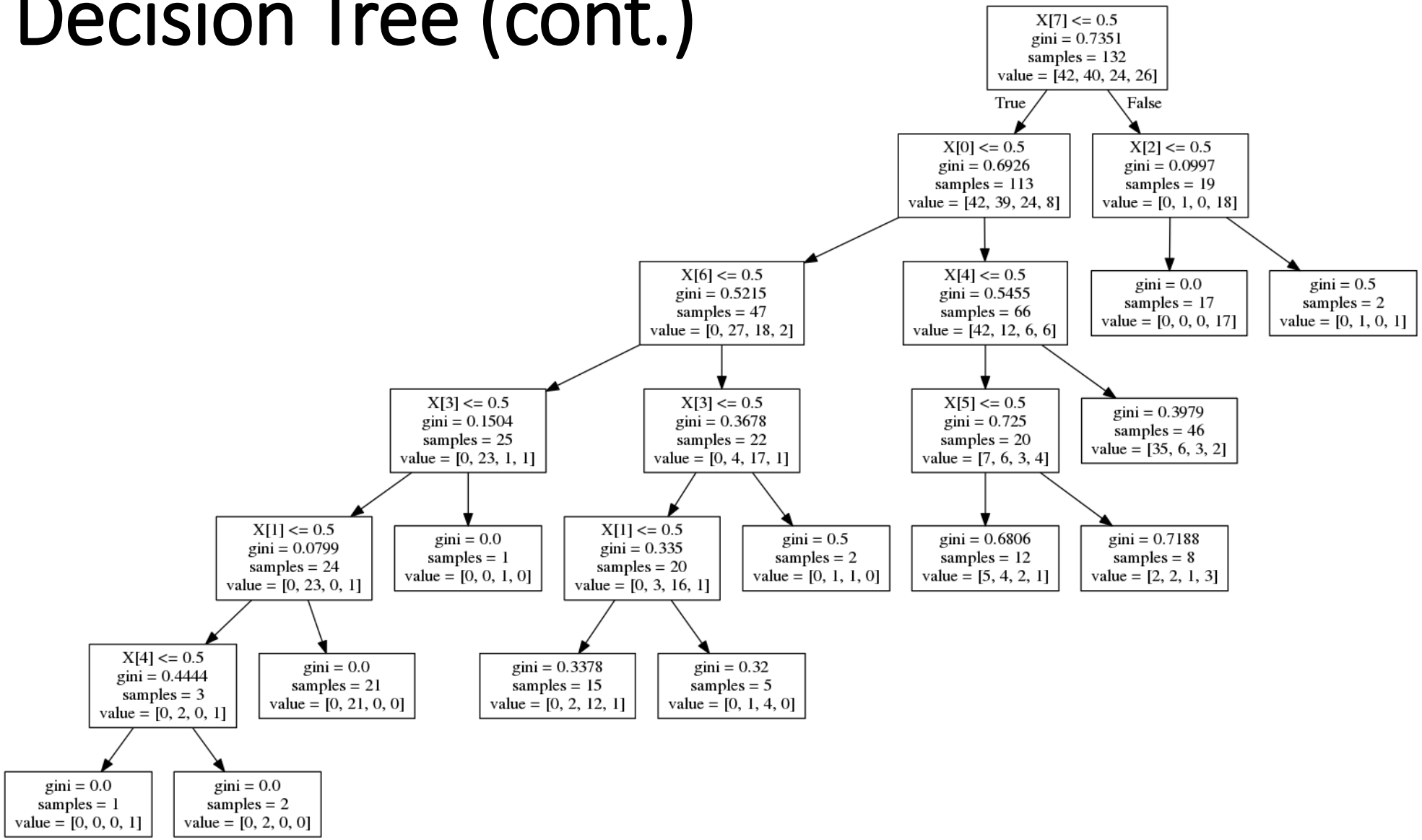


Image Taken from Introduction to Machine Learning by Datarocky

Decision Tree (cont.)

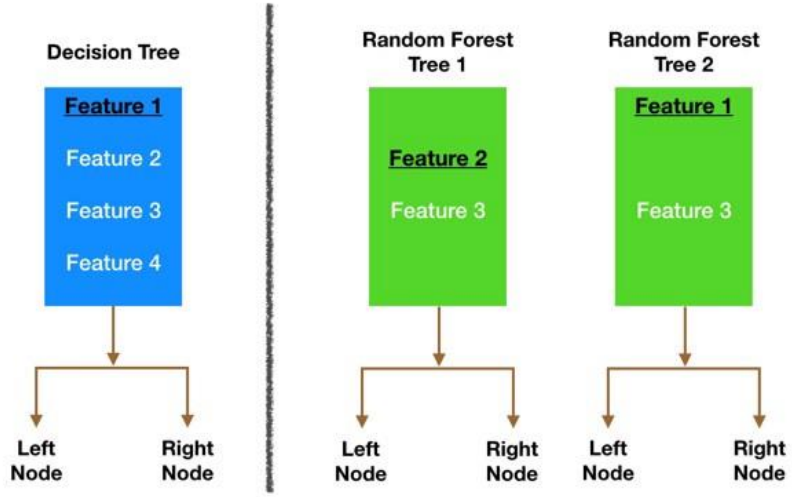


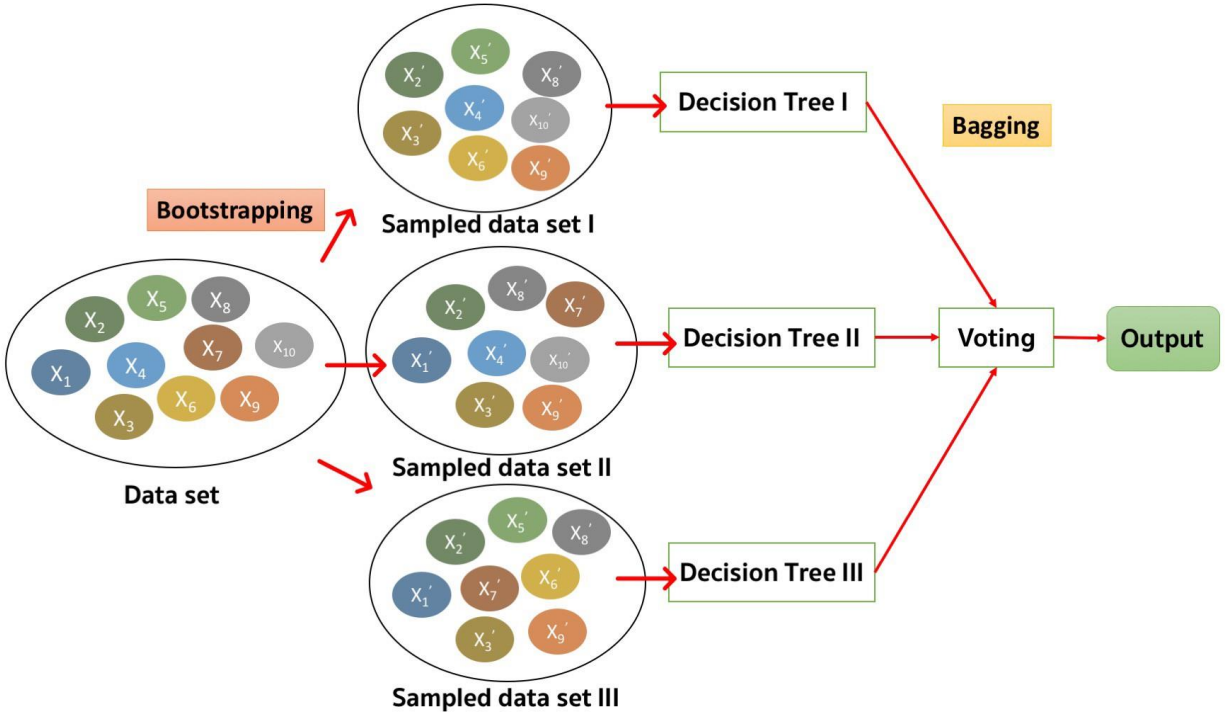
Decision Tree: Pros and Cons

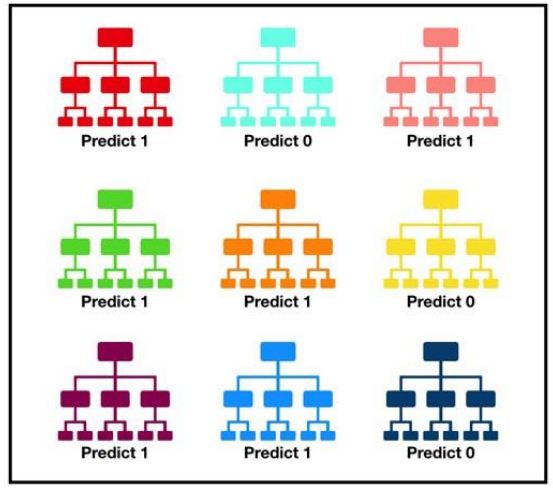
- Advantages
  - Easy to interpret the result
  - Easy to train
- Disadvantages
  - Easy to get overfitting (REMEMBER all previous solution, but cannot generalize for a new input)

Random Forest

- Ensemble based method
  - WISDOM of crowds
  - Grow trees from bootstrap samples (usually 33% of original data)







Tally: Six 1s and Three 0s
Prediction: 1

https://towardsdatascience.com/understanding-random-forest-58381e0602d2

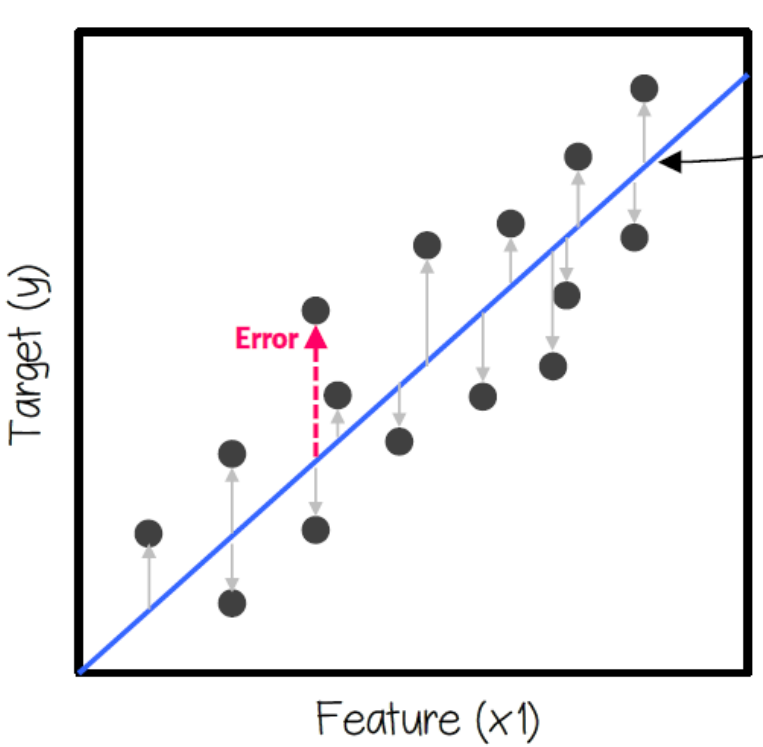
Random Forest: Pros and Cons

- Advantages
  - Improve overfitting issue
  - Easy to train
  - Can be used for both Classification and Regression problem

- Disadvantages
  - Can be slow to train for large dataset
  - Less explainable

Linear Regression

- Drawing a straight line that fit best with the given data points



We want to minimize RMSE (cost function)

Evaluation Metric

$$RMSE = \sqrt{mean\,(error^2)}$$

Image Taken from Introduction to Machine Learning by Datarocky

Regressions: Objective function

$$\sum_{i=1}^{M} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{M} \left(y_i - \sum_{j=0}^{p} w_j \times x_{ij} \right)^2$$

Linear Regression

$$\sum_{i=1}^{M} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{M} \left(y_i - \sum_{j=0}^{p} w_j \times x_{ij} \right)^2 + \lambda \sum_{j=0}^{p} w_j^2$$

Ridge regression

$$\sum_{i=1}^{M} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{M} \left(y_i - \sum_{j=0}^{p} w_j \times x_{ij} \right)^2 + \lambda \sum_{j=0}^{p} |w_j|$$

Lasso regression

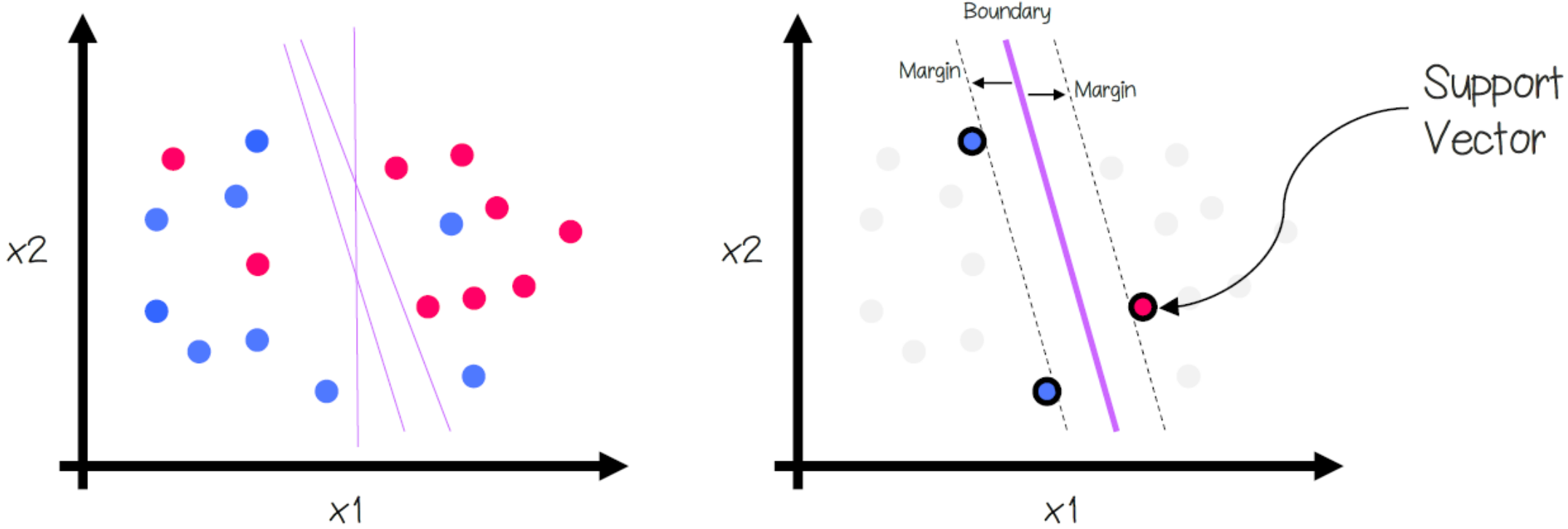
Colab

- Lecture
- https://colab.research.google.com/drive/1_R4A7OVSBESyg0M-IpF6GriJsSG70pVI?usp=sharing

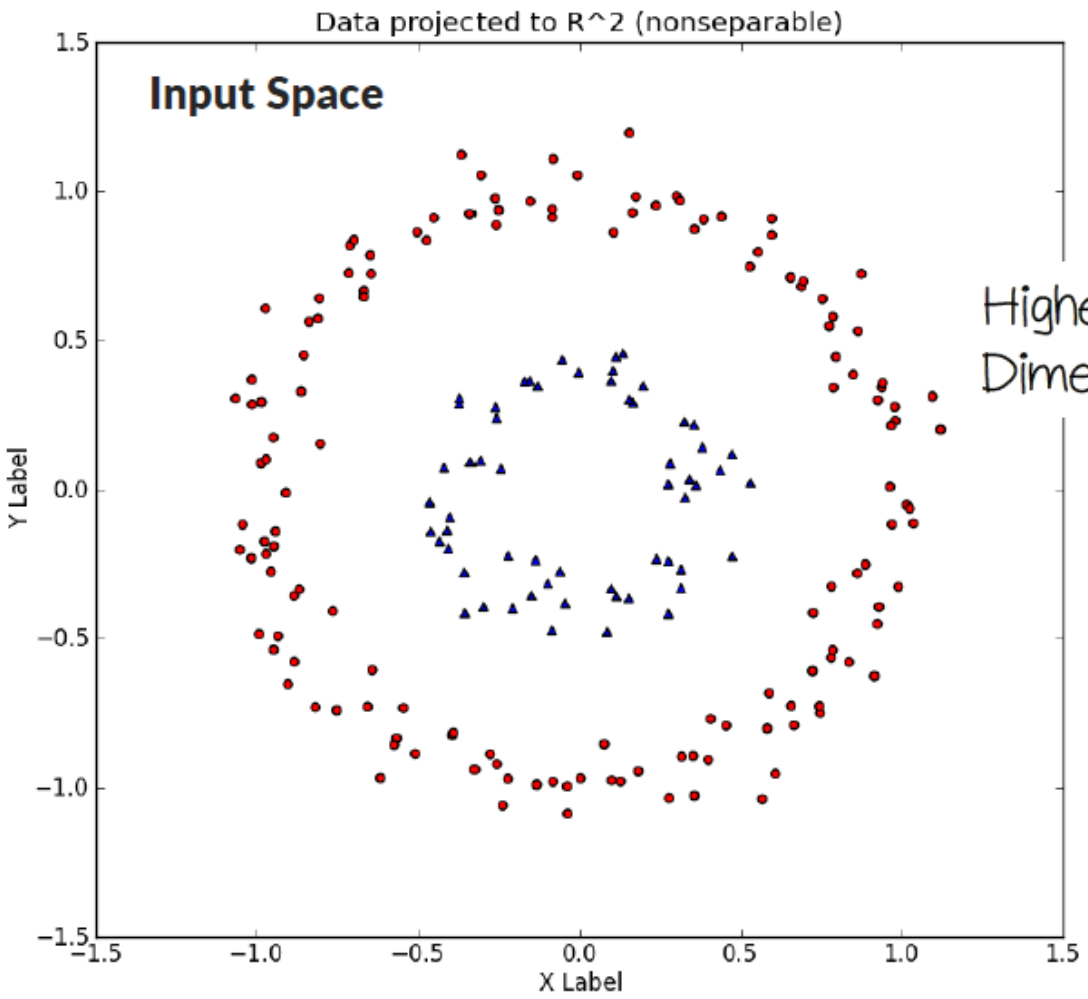
- Exercise
- https://colab.research.google.com/drive/1tJ_TeLo7_6MCliN1UQpblBvjzDR_gL0p?usp=sharing

Support Vector Machine

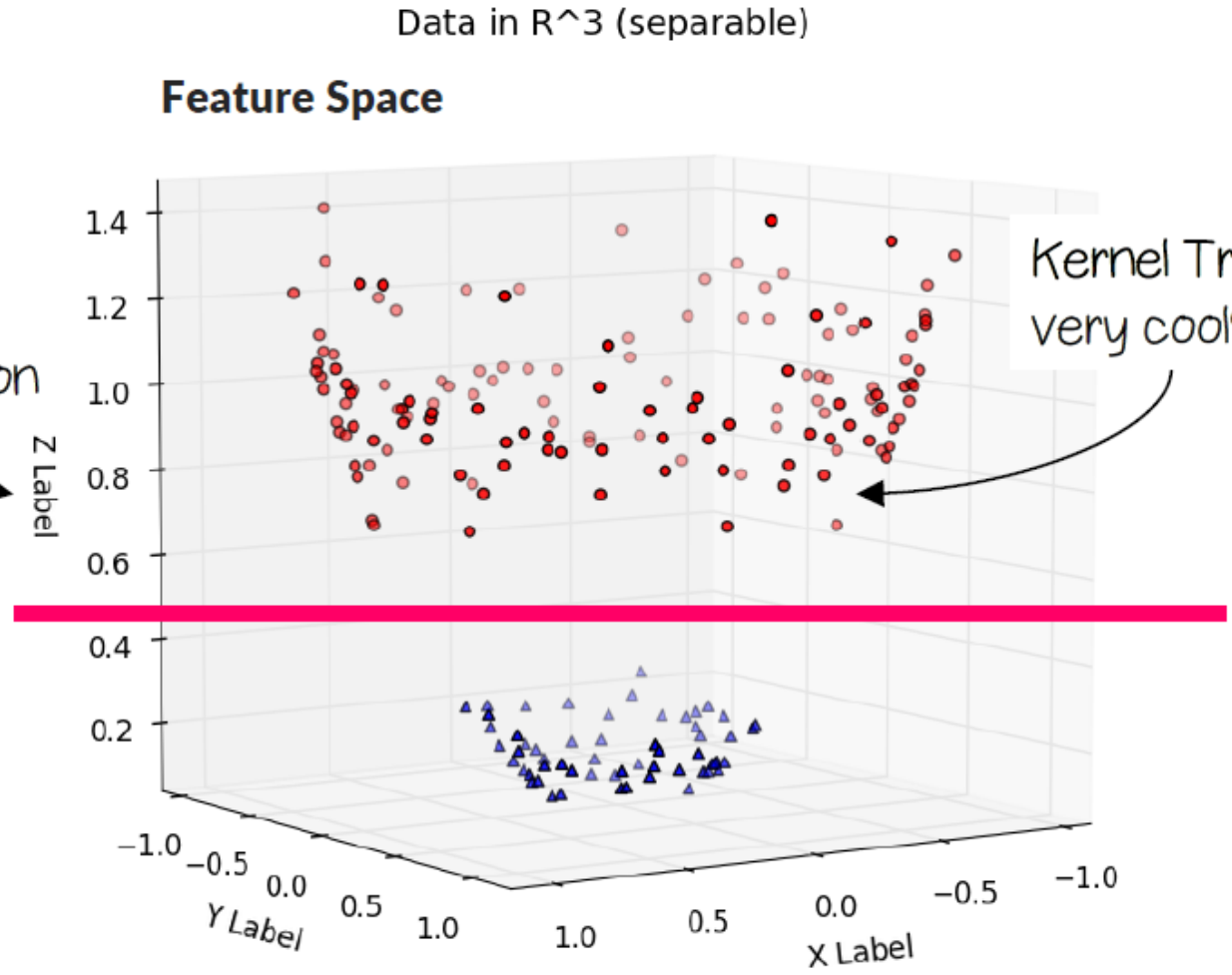
- Maximizing the margin from decision boundary



But can you divide this with a single straight line?

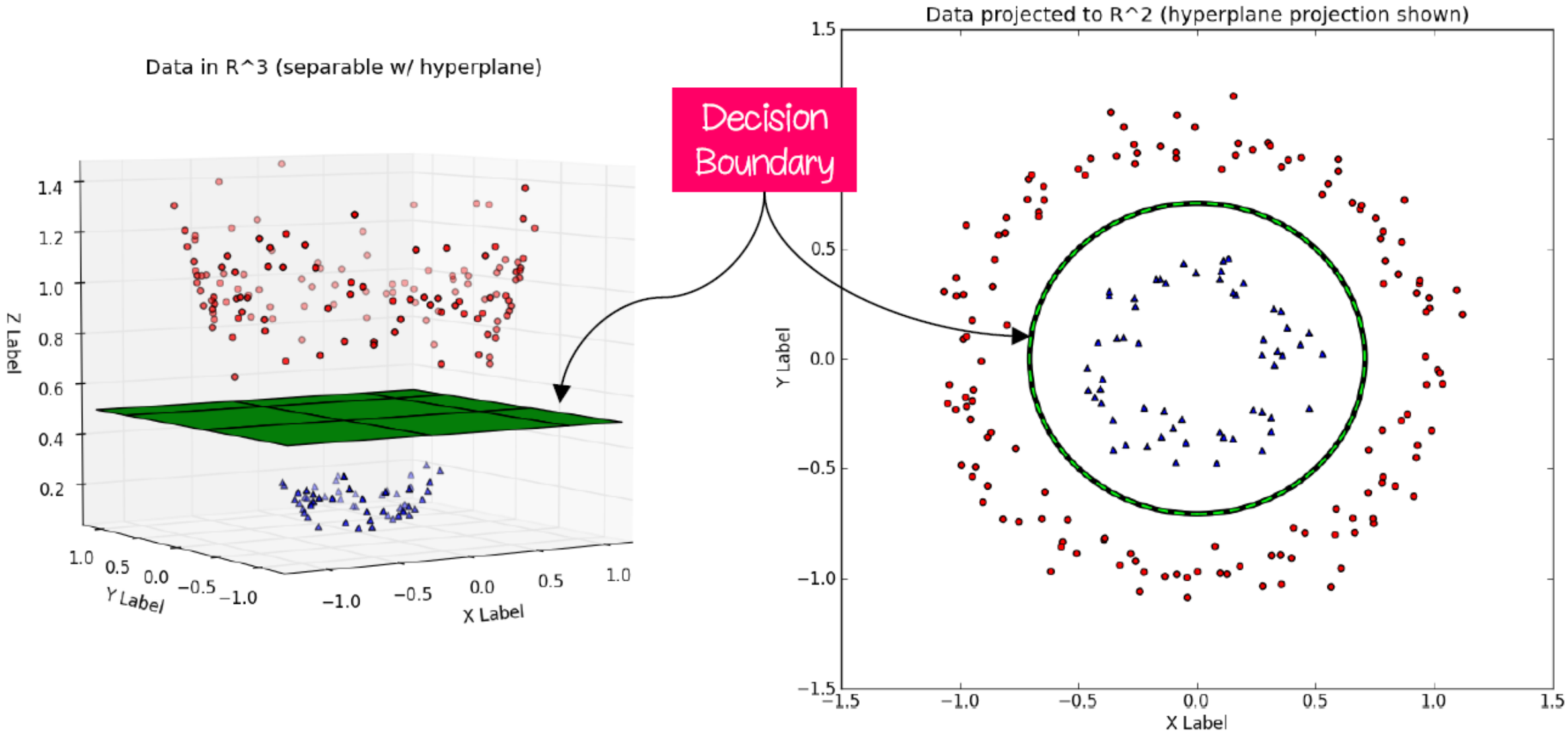


Higher Dimension



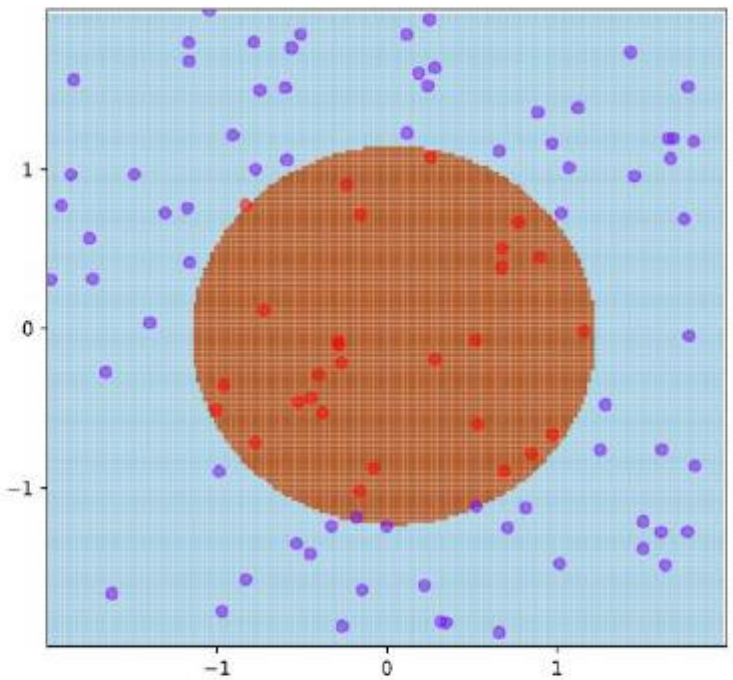
Kernel Trick, very cool?

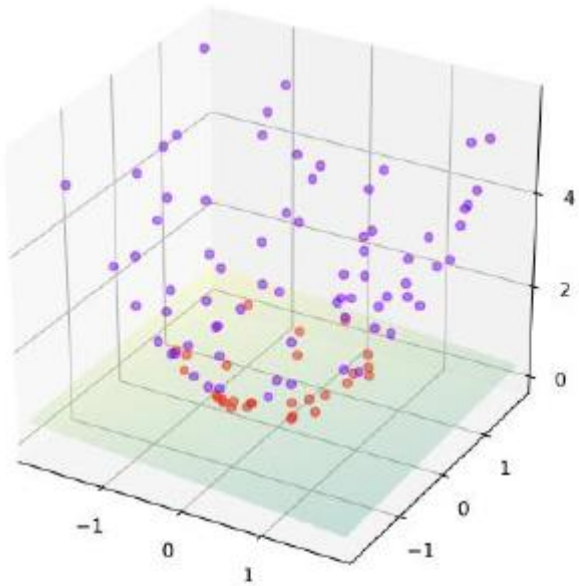
Impressive Result!!



Let's look closely at kernel trick

- *we have (x,y) coordinates*
- *we project a new higher dimension:* $(x, y, \underbrace{x^2 + y^2}_{3^{rd}\ dimension})$



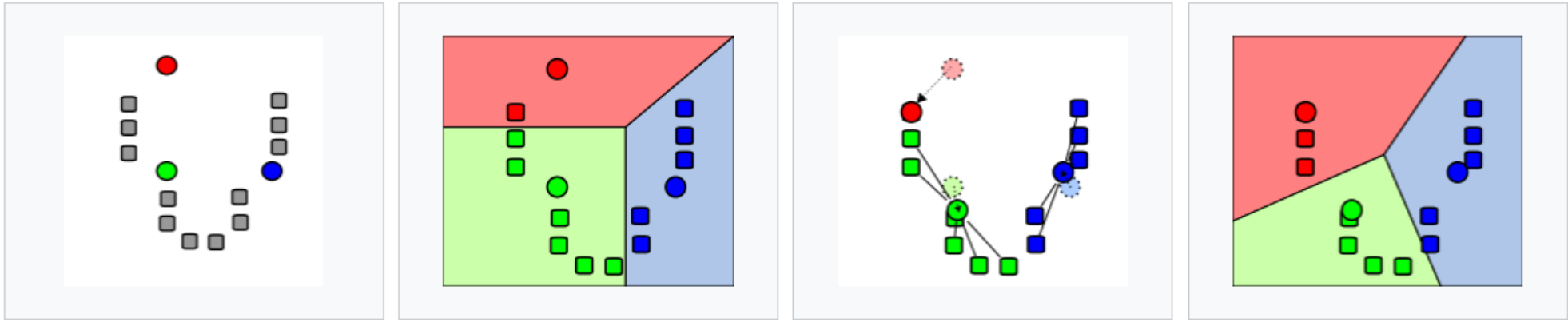


https://en.wikipedia.org/wiki/Kernel_method

K-Means (Clustering)

$$\arg\min_{\mathbf{S}} \sum_{i=1}^{k} \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 = \arg\min_{\mathbf{S}} \sum_{i=1}^{k} |S_i| \operatorname{Var} S_i$$

Demonstration of the standard algorithm

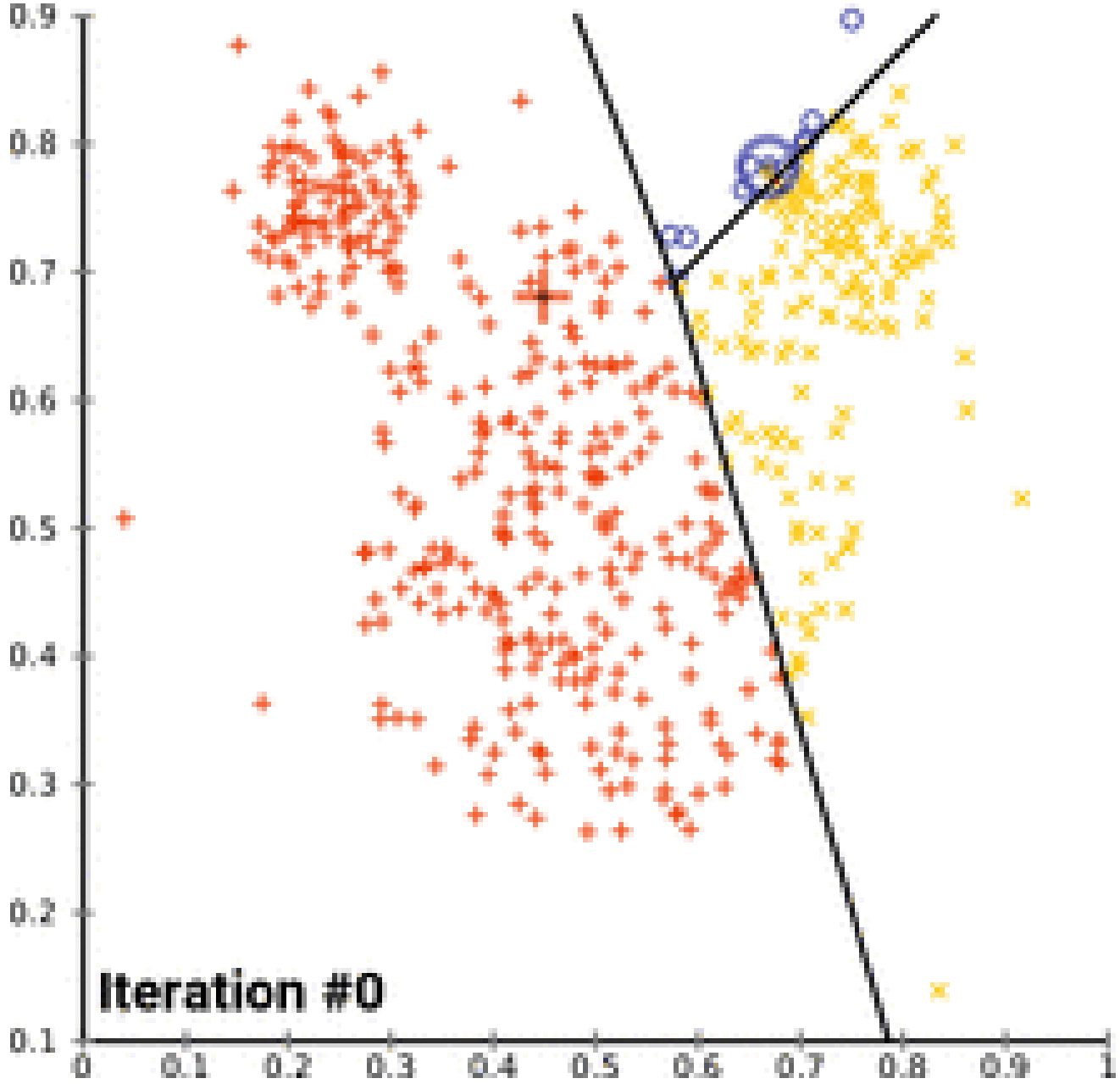


1. *k* initial "means" (in this case *k*=3) are randomly generated within the data domain (shown in color).

2. *k* clusters are created by associating every observation with the nearest mean. The partitions here represent the Voronoi diagram generated by the means.

3. The centroid of each of the *k* clusters becomes the new mean.

4. Steps 2 and 3 are repeated until convergence has been reached.


0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Iteration #0

Example of unsupervised algorithms

- https://towardsdatascience.com/the-5-clustering-algorithms-data-scientists-need-to-know-a36d136ef68

References

- Notes On Using **Data Science & Machine Learning**
  - https://chrisalbon.com/
- NLP
  - https://www.thainlp.org/pythainlp/tutorials/

- Machine Learning pipeline: Iris example (old version)

https://colab.research.google.com/drive/1TFTftpp1VpqhGfVfwkCCGqkwP_6acCIS?usp=sharing